Overview of Nata Supermarket

Nata Supermarket is a grocery retailer founded in 1972 that operates in Brendan, a city in Manitoba, Canada (Nata Supermarkets: Customer Analytics, 2023). The company supplies a wide range of products such as wine, fish, fruit, meat, gold, and confectionery. The business started as a family store, which expanded remarkably because it has historically prided itself on a customer-centered focus. In addition, the company pioneered data-driven inventory management through detailed transaction records, which helped them increase revenues and optimize product offerings.

Although Nata Supermarket adopted data practices early, it performed poorly in 2021 due to particular internal metrics compared to competitor growth rates (Nata Supermarkets: Customer Analytics, 2023). Based on the case study, Vina Verago has identified the two major problematic aspects of the company's performance:

1. Promotion targeting: Nata Supermarket has encountered difficulties in reaching and connecting to the right customers through targeted promotions.
2. Product inventory: Nata Supermarket has faced challenges in predicting customer preferences and behaviour, which led to poor inventory decisions.

On the other hand, the company's competitors like Walmart have been implementing more advanced data analytics initiatives. For instance, Walmart was the first one to formulate an analytics and machine learning division, which led them to optimize their business operations and extract meaningful insights through statistical models. In short, large competitors have experienced industry-leading growth through investment in innovative analytics and data-driven approaches.

Nata Supermarket has figured that this competitive disadvantage can be addressed by utilizing internal data from three years (2012-2014) on customer shopping habits (e.g., customer activity, purchase channels, spending behaviour) and customer demographic information. However, Vina Verago is now determining how this data can be leveraged effectively to improve internal metrics and help Nata Supermarket to be more competitive in an increasingly data-driven retail environment.

Analysis of Internal and External Factors

Strengths	Weaknesses
• **Established a good foundation of data collection and historical data:** Nata Supermarket has developed its internal dataset to collect relevant information on customer demographics and customer shopping habits in order to compete with data-driven competitors. The company is also a pioneer in using data to create informed decisions on product stocking levels. The three-year period dataset provides opportunities for trend analysis that one-time data collection cannot create. • **Detailed multichannel sales data:** Nata Supermarket has recorded various shopping statistics across different shopping channels (in-store, online, and catalogue) through the variables: NumWebPurchases, NumCatalogPurchases, NumStorePurchases, NumWebVisitsMonth This will allow Nata Supermarket to compare customer behaviour and average order values across multiple channels for different product categories and customer segments.	• **Underdeveloped data analytics and predictive capabilities:** Nata Supermarket has difficulties understanding and forecasting customer behaviours and preferences, which is a crucial data analytics initiative for annual reports that many competitors have taken. While Walmart has deployed predictive models for different business processes, such as anticipating checkout demand and determining the best form of product checkout, Nata has not caught up to similar capabilities. • **Limited campaign effectiveness:** Nata Supermarket has identified target promotions as a major metric that went subpar. This implies their inability to utilize data analytics to determine which customers should receive which promotional offers. This leads to wasting market expenditures and missed sales opportunities.

Opportunities	Threats
• **Customer segmentation implementation:** Nata Supermarket can segment customers into groups based on demographics, purchasing habits, and promotion engagement. As a result, it will lead to various positive outcomes: ○ Personalize marketing campaigns ○ Improve inventory management ○ Optimize pricing strategies ○ Enhance customer retention ○ Predict demand more accurately This will help Nata Supermarket to increase revenue and customer response rates and reduce stock outs or overstock issues. • **Predictive analytics implementation:** Nata Supermarket can utilize machine learning and predictive algorithms to open a lot of opportunities in terms of promotional performance, business efficiency, and customer engagement. For example, machine learning models can analyze customer purchasing behaviour and demographics in order to identify which customers are more likely to respond to particular deals. These models can be utilized for churn prediction, demand prediction, and cross-selling or upselling.	• **Competitors' data-driven and advanced analytics capabilities:** Walmart and other competitors have already developed sophisticated data-driven decision-making, potentially leaving Nata behind. A significant amount of investment and time is needed for technology, organizational change, and expertise for this implementation. This may position Nata Supermarket at a competitive disadvantage. • **Increasing Data Privacy Regulations:** There is growing scrutiny from consumers and regulators about how one's personal information is collected, stored, used, and protected. Canada's PIPEDA (Personal Information Protection and Electronic Documents Act) strictly imposes large penalties for non-compliance. When customers feel that their data is not secure, this may urge them to disengage from loyalty programs or promotions.

The SWOT analysis has effectively highlighted Nata Supermarket's current position in the competitive retail environment, which highlights both the strengths and weaknesses of utilizing data analytics. The strengths section emphasized the company's solid data foundation, which can be utilized to capitalize on opportunities that will help them improve and align with their competitors. On the other hand, the weaknesses accurately determine their inadequacy in analytics capabilities and campaign effectiveness, which are significant barriers to the company's performance growth. The opportunities section points out customer segmentation and predictive analytics implementation since it recognizes the transformative potential and will directly address the identified weaknesses in inventory management and promotion targeting. Ultimately, the threats section realistically emphasizes that Nata has been facing challenges and competitive disadvantages against data-mature competitors such as Walmart and the increasing complexity of data privacy regulations. The SWOT analysis lays out a comprehensive roadmap for where Nata Supermarket should prioritize its data analytics efforts in order to enhance its market position.

The Role of Data and Information in Supporting Business Processes

In today's modern digital economy, data and information are critical assets that streamline business processes, enabling organizations to improve efficiency, make data-driven decisions, and gain a competitive advantage in the market (Kraus, 2021). Businesses heavily depend on data to ensure that every aspect of the organization is aligned with their strategic objectives and goals. Aside from that, the dissemination of timely and accurate information cultivates better communication across teams and departments. Data and information provide value in business processes in various ways:

1. Improving Efficiency
    - According to Ladley and Redman (2020), insights that are derived from data assist organizations in optimizing and automating business operations, which reduces manual efforts and redundancies. With that, data and information help determine bottlenecks in workflows such as manual approval processes, limited resources, and inefficient communication. Real-time analytics enables businesses to monitor key performance indicators (e.g., website traffic, user engagement, cash flow), and enhance operational decision-making. It will allow

businesses to evaluate their progress toward their goals, whether short-term or long-term, and make adjustments as needed. Ultimately, access to data and information empowers businesses to make fast and informed decisions. For instance, predictive analytics allow organizations to manage inventory efficiently, forecast demand, and optimize supply chains.

2. Identifying Opportunities
    - Conducting market data analysis, involving economic conditions, industry developments, and customer preferences, assists businesses in determining relevant and emerging trends in the industry, as well as new market segments. Analytics can uncover untapped markets, innovative product ideas, and customer needs that can give the company a competitive edge. Customer data can be turned into information that will enable personalized marketing strategies, which result in increased brand loyalty and engagement. Competitive analysis helps organizations figure out gaps in the market, wherein they can spot unaddressed customer problems or pain points that other competitors tend to overlook. Data and information about competitors' successes and failures drive the business to develop creative approaches to better accommodate customer needs.
3. Mitigating Risks
    - Data analytics provides substantial support for risk assessment, which allows organizations to detect potentially fraudulent behaviour or compliance issues. For example, unusual transaction amounts in bank systems may indicate fraud. Also, real-time data on monitoring systems helps organizations to identify and respond swiftly to cybersecurity threats, which reduces the likelihood of downtime or costly data breaches. With data and information, financial risk management can benefit from it. Predictive models, like credit risk prediction, examine customer credit histories, economic trends, and income levels, which evaluate the likelihood of loan defaults. Instead of being reactive to these kinds of incidents, data and information allow proactive risk management through immediate prediction and acknowledgment of threats ahead of time. Therefore, this allows risk-aware decisions and reduces financial losses that safeguard the business.

4. Enhancing Customer Experiences
    - Through consolidation and the process of customer data from multiple channels in real-time, the implementation of a customer data platform (CDP) can be done (Garg, 2022). A CDP equips the organization to monitor customer engagement and interactions across various channels, including mobile applications, websites, in-store systems, and interactions with business partner systems. Data centralization helps organizations improve customer experience through personalized recommendations and search results. In addition, an AI-driven recommendation engine can be built using user data to analyse past interactions, preferences, and browsing behaviour, which enables businesses to carry out highly tailored and relevant suggestions that enhance customer satisfaction and boost conversions. Another example is that a traditional Customer Relationship Management System (CRM) monitors basic customer behaviour, however, it may miss the key emotional components of customers.

The Critical Role of Data Quality and Security in Organisational Value Creation

Data quality has become increasingly critical to organisations as they heavily rely on data for their procedures. Data quality is defined as the accuracy, reliability, consistency, and completeness of data (GeeksforGeeks, 2024). The higher the quality of data, the more likely the data is free from inaccuracies and errors, which makes it ideal for decision-making and analysis. Data quality is prioritized in organisations in order to guarantee that their information assets satisfy the needed standards and contribute effectively to decision-making and business operations. An organisation with high data quality is much more advantageous than organisations that does not have it. Over the past decades, there has been an exponential growth of data or big data being produced and stored due to the Internet of Things, Artificial Intelligence, and edge computing. With that, maintaining the accuracy and organisation of data has become a heavier task. Therefore, each organisation must have Master Data Management (MDM). This is a technology used to manage organisational data across the enterprise by consolidating and unifying data assets from multiple sources, such as customer information, location data, and product details (Mucci, 2024). The data must ensure correctness, consistency, and be well-maintained throughout the systems. Building

on that, it is important to hire data stewards and set stricter rules to maintain high data quality. Business dashboards with poor data quality can lead to disastrous repercussions, especially in the field of healthcare, where the stakes are incredibly high. It can lead to misguided strategies, inaccurate conclusions, and financial losses. GeeksforGeeks (2024) and Kaufmann (2024) highlight the profound value of data quality to an organisation:

- It is capable of developing key performance indicators (KPIs) that evaluate the effectiveness of projects.
- It allows teams to pinpoint the aspects of operational workflow failures. It streamlines operational management processes and minimizes inefficiencies. With that, it helps develop new products and gain a competitive advantage.
- It lowers the legal and compliance risks since it adheres to data quality standards. This prevents penalties and reputational damage.
- It reduces the likelihood of mistakes, boosts productivity, and enables informed and quick decision-making.
- It lowers the need for revisions, which is cost-saving.
- It instils confidence, promotes trust, and improves the organisation's credibility with its stakeholders.
- Consistent and accurate data improves customer satisfaction, fostering trust and loyalty, and maintaining positive customer relationships. This enhances brand reputation and customer retention. Inaccurate data, like incorrect billing information, is likely to lead to seeking alternatives and being dissatisfied with the service.

On the other hand, data security is equally important to data quality. Data security is a proactive protection of digital information from damage, theft, and corruption (Houghton, 2024). The real value of data is only realized when it is secured. The value of data security lies in the following: (1) Protect critical data against data breaches, (2) Prevent data and financial loss, (3) Support business continuity, and (4) Ensure regulatory compliance. In today's digital landscape, data security is not just a compliance requirement, but it is the key driver of innovation, trust, and sustainable long-term business success, especially in this competitive and dynamic environment. Organisations that incorporate data security into their operations not only safeguard information but also allow data-driven

decision-making. Houghton's (2024) and Sutevski's (2023) articles emphasize the critical importance of data security to an organisation:

- Protecting data integrity and reliability
- Enhancing brand reputation and customer trust
- Risk mitigation and regulatory compliance
- Allowing competitive advantage and data-driven innovation
- Operational continuity and resilience

Inadequate security measures enable the data to be susceptible to unauthorized access, corruption, and cyber threats, which compromises the integrity (Gonsalves, 2024). Implementing data security through different technologies such as encryption, real-time monitoring, and access controls ensures that organisations work with unaltered and correct data, which leads to reliable decisions. Moreover, Data Loss Prevention (DLP) technologies help protect against intentional or accidental data leaks. A proactive response to data security, including disaster recovery plans and cybersecurity strategies, guarantees business continuity even amidst cyber incidents. Since customers entrust businesses with confidential and sensitive information, data breaches and cyber-attacks may tarnish the organisation's reputation, leading to loss of trust, legal consequences, and customer churn (Sutevski, 2023). Data security reinforces organisational's commitment to compliance, privacy, transparency, improving brand value, and strengthening customer relationships. The implementation of robust cybersecurity policies and proper training of employees on best security practices ensures that internal threats are addressed, which cultivates a culture of security awareness.

It is now imperative for businesses to comply with data protection laws such as CCPA, GDPR, and industry-specific regulations (Houghton, 2024). Without data security, it exposes organisations to legal liabilities, financial penalties, and operational disruptions. This highlights the importance of reinforcing robust security frameworks that guarantee compliance with regulatory requirements, which improves overall corporate resilience. Ultimately, businesses must stay updated with evolving security regulations and cultivate proactive compliance techniques to prevent expensive legal repercussions. Data security unlocks the full potential of organisations toward advanced analytics, artificial intelligence, and business intelligence tools (Avakov, 2024). This is because it helps organisations to confidently and safely navigate emerging new technologies and formulate

innovative solutions that lead to business growth. Ultimately, a firm security posture cultivates an environment where data can be utilized strategically to gain a competitive advantage and generate new revenue streams.

Value of Data and Information to Individuals and Organisations in Nata Supermarkets

	Value to Individuals	Value to Organisations
Strategic Value	**For Customers:** - Based on demographic data and previous purchase behaviour, customers are provided with personalized shopping experiences. - Product recommendations are very relevant, which eases the access of customers to their desired items. **For Employees:** - Data-driven performance tracking helps them understand both strengths and weaknesses accurately. - It increases workflow efficiency since it automates repetitive tasks.	- Nata Supermarkets can resolve disparity with competitors, like Walmart, that utilize data for market domination. - Data and Information assist the company in focusing on factual insights rather than intuition-based decisions. - With data analysis, it provides a lot of market expansion opportunities since it identifies trends, customer needs, and potential spots for new branches or stores.
Operational Value	**For Customers:** - Personal shopping preferences and habits are matched through effective promotions. - It will reduce frustrations among customers due to improved inventory management. - Through centralized customer service tracking, returns and complaints are resolved easily. **For Employees:**	- Data and information allow precise customer segmentation, which results in higher conversion rates and engagement. - Real-time data on stock levels helps avoid shortages and overstocking, which improves overall stocking

	- Through customer traffic data and sales trends, resource allocation and scheduling are improved. - With customer feedback and interaction data, the training programs are much more effective.	efficiency.
Customer Understanding	**For Customers:** - To create a sense of exclusivity, loyalty rewards can be done based on shopping behaviour. - Product assortments and tailored recommendations based on individual purchase history. **For Employees:** - They can provide better assistance to customers since they gain data-driven insights into customers' past purchases and preferences. - Data and information enhance personalized interaction and service quality.	- Nata Supermarket can determine seasonal variations, product popularity, and purchasing trends. - Different customer groups can be segmented, such as high-value customers and low-value customers. - Customer Lifetime Value (CLV) helps allocate resources effectively.

Impact of Using Data and Information to Support Nata Supermarkets

Data is a transformational asset for success, and it is no longer an optional asset in this modern retail landscape. The case study of Nata Supermarkets uncovers how data and information can impact the performance of organisations in several dimensions. The organisation finds itself at a turning point where proper and effective data utilization could alter the outcomes of the business. Nata Supermarkets started with a data-driven approach by using manual ledgers to monitor transactional data. This primitive analytics system initially drove the company its successful expansion to 37 stores in Canada. However, Nata was unable to modernize its data approach, which led to underperforming against competitors, poorly targeted promotions, limited understanding and prediction of customer behaviour, and suboptimal inventory management. The extensive

customer data they have gathered is a significant and untapped opportunity that could change their overall performance.

	Specific Applications	**Data Categories and Expected Impact**
Strategic Decision-Making	Market Positioning	Education, Income, and Spending patterns help determine the most profitable customer groups and positioning opportunities.
	Competitive Response	All the data categories bridge the gap with other data-driven competitors, which can potentially gain "industry-leading growth."
	Resource Allocation	NumWebPurchases, NumCatalogPurchases, NumStorePurchases help optimize investments in sales channels such as online, catalogue, and in-store.
Tactical Decision-Making	Promotion Effectiveness	AcceptedCmp1-5 and Demographics help develop target promotions to receptive customer groups.
	Product Mix Optimization	Spending on categories such as MntWines, MntFruits, etc, optimizes stocking and merchandising decisions.
	Channel Strategy	Purchase Channels and NumWebVisitsMonth help create a cohesive omnichannel strategy, such as a comparison of online vs in-store sales.

Operational Efficiency	Inventory Management	Recency and Product categories spending reduce wastage and stock outs by improving the predictions.
	Staff Scheduling	Purchase timing data gives insights into predicted customer demand, which optimizes workforce scheduling.
	Supply Chain Optimization	Purchase patterns assist in informing supplier relationships and ordering schedules.
Customer Experience	Personalization	Purchase patterns and Demographics significantly allow personalized product recommendations and promotions
	Complaint Resolution	Complain Field and Demographics proactively acknowledges customer satisfaction issues and enhances service quality.
	Loyalty Enhancement	Purchase patterns and Recency help strengthen customer retention programs and bring back engagement in at-risk customers.
Performance Evaluation	Campaign ROI	Spending and AcceptedCmp fields measure the campaign's effectiveness, beyond simple acceptance rates.
	Customer Lifetime Value	Dt_Customer, Spending, and Demographics help calculate Customer Lifetime Value as

		time goes by.
	Store-Level Comparisons	Location data determines top-performing stores and incorporates successful practices.
Business Model Innovation	Subscription Services	Purchase Frequency aids in formulating subscription-based revenue streams.
	New Product Categories	Demographics and Category Spending help expand product offerings and determine unaddressed customer needs.
	Digital Transformation	NumWebPurchases and NumWebVisitsMonth can improve e-commerce capabilities through online investment priorities.

Nata Supermarket can fully capitalize on the benefits of data analytics if it addresses the challenges systematically:

1. The organisation should implement modern analytics capabilities such as machine learning and predictive algorithms, according to the case study. However, this would need investment in specialized data science talent or extensive training of existing staff to effectively handle these emerging technologies.
2. Vina Verago highlighted data visualization tools as a rising technology of interest, indicating promising aspects for investment. Data visualisation transforms complex data into understandable and accessible insights that are easy to act upon.
3. Nata Supermarket will undergo a cultural shift toward data-driven decision-making across all aspects of the business. This means they have to embrace modern analytics capabilities and move away from intuition-based and outdated approaches that they have been using in the past.

4. The organisation must integrate robust data quality management practices in order to ensure the completeness, accuracy, and timeliness of the data it collects. Sophisticated analytics tools will be useless and lead to misleading results when the data is unreliable or uncleaned.

Data and Information within Organisations

Data Generation and Collection

1. Surveys
    - Surveys collect structured data or feedback directly from target respondents on particular topics or research (Qualtrics, 2018). It involves questions related to the respondents' opinions, experiences, preferences, and behaviours in order to gain insights into the particular population. The data collected requires the use of statistical analysis and software to identify correlations, interpret results, and gain meaningful insights. There is a wide range of online survey platforms that can be used, such as HubSpot, Typeform, Qualtrics, and many more. Examples of surveys include:
        i. Customer Satisfaction Surveys – It is a questionnaire used by businesses to measure customers' feelings or satisfaction level about a service or product (Bhasin, 2015).
        ii. Market Research Surveys – It is a questionnaire used by businesses to understand the market better, such as competitive landscape, customer preferences, and behaviours, which helps make informed decisions about product development, content creation, and marketing strategies (Pollfish, 2023).
        iii. Employee Feedback Surveys – It is a questionnaire used by businesses to gain feedback about an employee's job satisfaction, career development, and work environment, which are used to improve employee experience, company culture, and employee retention (PeopleHR, 2024).
2. Customer Inputs
    - Data collection is not only limited to surveys, customers provide crucial data through different interactions. It is defined as the

recommendations, enhancement requests, suggestions, or other inputs by customers in relation to the operation or functionality of the service or product (Ulwick, 2002). Customer inputs can be gathered from various sources:

i. Product Reviews and Ratings – It is the positive or negative comments from customers who have bought and used the product, whether qualitative or quantitative.
ii. Registration Information – It is the information provided by the customers during account creation, such as demographic, contact details, and preferences.
iii. Support Tickets – It is a piece of documentation between the customer and the customer support team (Fuchs, 2019). A ticket is generated when a customer submits a query to customer service, requesting assistance with a certain problem.
iv. Social Media Interactions – It is described as the comments, reactions, shares, and direct messages made by users to a certain post or platform. It can be used for conducting user sentiment analysis.

3. Web Scraping
    - It is an automated method to collect large amounts of data from a website systematically and quickly (GeeksforGeeks, 2020). The data collected is usually in HTML format and unstructured, which is transformed into structured data in a database or spreadsheet for different applications. Therefore, it requires extensive structuring and cleaning before data analysis due to the inconsistencies in formatting. This is usually done for the purpose of conducting market research and training machine learning algorithms. Examples of web scraping include:
        i. Competitor Price Monitoring – It automatically collects the pricing information through competitors' websites.
        ii. Product Information Aggregation – It is the collection of features and specifications of products on a particular manufacturer's website.
        iii. Content Aggregation – It gathers relevant web content such as research, articles, and publications.
4. IoT Devices and Sensors

- Internet of Things (IoT) is defined as the network interconnection of vehicles, applications, and physical devices, embedded with software, sensors, and connectivity capabilities, which allows generating continuous streams of operational data (Srivastava, 2023). These devices gather and exchange data, which enables them to communicate and interact with one another in the digital world. Examples of IoT devices and sensors include:
    i. Wearables – Devices like smartwatches, fitness trackers, and health monitoring devices involve functionalities like health monitoring, personalized notifications, and activity tracking, which empower individuals to sustain a healthy lifestyle (Hosseini, 2023).
    ii. Barcode scanner or sensor – It is defined as an optical device, consisting of a light source and a lens, that scans and converts the data into a standard format that can be processed by a computer (GlobalSpec, 2015). This is widely utilized by various industries for managing sales, tracking inventory, and automation data entry processes.
    iii. Beacons – It refers to the hardware sensors in various areas of a retail store or a site (Amrita, 2024). This is deployed to track the number of visitors or foot traffic through body heat and movement.

5. Transaction Records
    - It is an electronic or written record of information relating to an Account or Instruction, which is one of the most structured and reliable data sources for businesses (Chen, 2024). It builds a foundation for cash flow monitoring, revenue analysis, inventory management, and identification of customer purchase patterns. The data collected is typically processed through Enterprise Resource Planning (ERP) systems, data warehouses, and accounting systems. Examples of transaction systems include:
        i. Point-of-Sale Systems (POS) – It is a software and hardware system, typically a cash register, that helps retail stores to make sales, accept payments, and check out customers (Shopify, 2022). However, in this digital landscape, POS systems can be

software-based and can be utilized on any tablet or smartphone.

ii. Financial Transactions – It is defined as the exchange of monetary value, financial assets, goods, or services for payment between a buyer and seller (Datta, 2024). Examples of financial transactions include withdrawals, transfers, deposits, and recurring payments.

iii. Warehouse Management System (WMS) – It refers to the different movements or flow of stock within and beyond a business (Reid, 2025). Movement inventory can be divided into categories, such as inbound, outbound, and internal movements. It monitors stock transfers, receipts, and depletions.

Various tools can be used to manipulate raw data at different levels in order to make it more readable, useful, and suitable for analysis. According to Astera (2020), the popular data manipulation tools are as follows:

- Spreadsheet Softwares
  - Microsoft Excel and Google Sheets offer various capabilities that are sufficient for basic data manipulation, such as sorting, filtering, and formulating simple calculations (Reza, 2022). Aside from these, it gives spreadsheet features such as charts, pivot tables, conditional formatting, and collaboration with other users.
- Statistical Softwares
  - SAS and SPSS provide advanced analytical and statistical tools that are widely utilized for business, research, and academic (Pedamkar, 2018). It supports various statistical methods such as descriptive statistics, ANOVA, and regression, which can effectively handle large datasets.
- Programming Languages
  - Python, R, and Julia provide powerful data handling libraries such as pandas and Numpy, dplyr and data.table, and DataFrames.jl and JuliaDB, respectively (Mohmad, 2024). It offers capabilities such as grouping, filtering, and summarizing data effectively. In addition, it is flexible in terms of both

structured (data frames or tables) and unstructured data (text files or raw data).

- Database Management Systems (DBMS)
    - MySQL and PostgreSQL are both relational databases that structure data into tables that support querying (IBM Cloud Education, 2024). It uses Structured Query Language (SQL) for data manipulation tasks like INSERT, SELECT, UPDATE, and DELETE operations. Ultimately, it provides a vast range of data types, including text, Booleans, and dates, for a standardized format and manipulation.
- Business Intelligence (BI) Tools
    - Tableau and Power BI are powerful tools, primarily used for advanced dashboard creation and interactive visualizations (Edmond and Crabtree, 2022). It features seamless data integration and connectivity, wherein it connects to various sources of data such as web services and cloud-based databases.

Legal and Ethical Considerations

The incorporation of data and information to support business processes has exponentially revolutionized business operations, customer engagement, and decision-making in today's digital landscape. However, this heavy reliance on data introduces important social, legal, and ethical considerations that organisations must address to ensure that data is leveraged responsibly. Nata Supermarkets presents the crucial transformation in modern business strategy, which is the change from intuition-based data to data-driven decision-making. This evolution mirrors competitive trends and broader technological advancements:

- Predictive analytics and machine learning algorithms allow sophisticated customer insights.
- Access to affordable computing power makes advanced analytics available to a broader audience.
- Data collection became a strategic asset for organisations.
- A major competitor, Walmart, has pioneered advanced data analytics in retail, which has let it gain a competitive advantage in aspects of:
    - Promotional targeting

- Inventory management
- Customer segmentation
- Personalized shopping experiences

Social Implications

The collection of large amounts of personal data increases concerns about one's privacy. According to Anant (2020), consumers are increasingly cautious about sharing personal data due to the prevalence of data breaches and misuse in this digital age, which can result in severe consequences such as identity theft, financial loss, and reputational damage. Most of the companies utilize personal data for customer insights, targeted advertising, and service improvements. However, consumers frequently feel wary and lack control over how their information is used, stored, and shared. Harrell (2024) found that 12% of people aged 16 or older in 2021 experienced a data breach. Also, victims of identity theft were twice as likely to find out their personal information had been misused, compared to those non-victims.

Another social implication is the digital divide. Most businesses, organisations, and potentially Nata Supermarkets, have shifted to data-driven decision-making, online platforms, and digital tools to operate efficiency, however, some individuals do not have equal access to these resources or struggle to participate in this digital revolution, especially for low-income individuals, marginalized groups, or rural communities (Sanders and Scanlon, 2021). For example, a person without Internet access may find it difficult to apply for a job online or for remote work opportunities, which limits their employability and economic opportunities.

Furthermore, bias and discrimination impact business processes (Sap, 2024; Chapman University AI Hub, 2025). Nowadays, many companies utilize AI-driven recruitment tools for screening job candidates. Through the utilization of AI, it may unfairly favour particular demographics over others if the AI is trained on biased hiring data. For instance, Amazon got rid of its AI hiring tool since it discriminates against female candidates, wherein it favours resumes of male candidates. This raises concerns for women, minorities, and individuals with employment gaps, facing unequal and unfair rejections. Similarly, discrimination in financial services occurs. Banks and lenders depend on AI-based credit scoring to deny or approve loans for clients. However, it may unfairly penalize individuals from low-income

backgrounds or minority communities due to the historically biased financial data on which the AI model was trained. If bias and discrimination are embedded in business processes, they reinforce economic and social disparities, leading to adverse impacts on employment. For instance, the Facebook-Cambridge Analytica scandal uncovered how the utilization of unauthorized data could lead to manipulation of political campaigns, which causes widespread distrust and fake news on social media platforms (Bareebe, 2022). Uber's increased pricing algorithm has been backlashed for demonstrating inequalities by charging higher fares in low-income areas (Hall, Kendrick, and Nosko, 2015).

Data-driven decision-making in Nata Supermarkets and other businesses can also raise ethical concerns within customer marketing. Due to the usage of predictive analytics and behavioural data to influence consumer decisions, businesses sometimes push unnecessary purchases and excessive spending. Therefore, this may raise concern regarding consumer autonomy and ethical advertising practices.

Legal Implications and Regulatory Frameworks

Nata Supermarkets and other businesses collect and analyse large amounts of customer data, which is why compliance with data protection regulations is essential, particularly in a Canadian context (PIPEDA). General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA) are data protection laws and legal requirements regarding how businesses collect, use, and disclose personal information (Sarwar, 2024; Bonta, 2018). Companies must acquire explicit user consent beforehand, and must inform users about:

- What data is being collected
- How it will be used
- Whether it will be shared with third parties

These regulations ensure that individuals have the right to control their personal information. However, if organisations exploit or misuse data, it can result in identity theft, privacy violations, and surveillance concerns. Therefore, national authorities asses hefty fines for particular data protection violations in accordance with the UK/EU GDPR (Intersoft Consulting, 2021).

- Under GDPR, companies can be penalized up to €20 million or 4% of their global turnover.
- CCPA violations can result in penalties of $7,500 per intentional violation and $2,500 per unintentional violation.

Failure to comply will hold businesses legally accountable, which makes them liable for the threats and subject to regulatory penalties or lawsuits for the resulting outcomes. With a rapid increase in cybersecurity threats, GDPR forces organisations to implement robust security measures to protect user information from breaches. On January 21, 2019, a notable GDPR violation occurred against the company GOOGLE LLC (EDPB, 2019). They faced a $57 million fine for not being transparent and the inadequacy of information regarding user consent for the processing of data. The key principles of GDPR include (GDPR Summary, 2020):

1. Lawfulness, Fairness, and Transparency
2. Purpose Limitation
3. Data Minimization
4. Accuracy
5. Storage Limitation
6. Integrity and Confidentiality

GDPR offers individuals with important rights, including:

- Right to access or request their personal data records.
- Right to rectification or correction of inaccurate data.
- Right to be forgotten or request data erasure.
- Right to restrict processing or object to data use in particular situations.

The Data Privacy Act (DPA) of 2012 is the Philippines' data protection law:

- The National Privacy Commission (NPC) is the regulatory body that is assigned to enforce compliance and examine data breaches.
- Organisations must ensure legitimate purpose, transparency, and proportionality when processing data.
- Rights of Data Subjects, such as the right to be informed, access, correct, and erase personal data.

Another real-world example is the Philippines' COMELEC website suffered a data breach by two hacker groups which exposing the personal information of 55 million registered voters (Masaga, 2023). This was one of the largest data breaches in history, also known as 'Comeleak'.

GDPR, CCPA, and DPA have set strong precedents for data privacy. However, Canadian companies, like Nata Supermarket, must strictly adhere to PIPEDA — the Personal Information Protection and Electronic Documents Act, Canada's federal privacy law. In line with this, Nata supermarket can ensure legal compliance by aligning its data handling practices with PIPEDA's 10 Fair Information Principles (Office of the Privacy Commissioner of Canada, 2024), which are as follows:

1. Accountability – Nata Supermarket should designate individuals responsible for ensuring effective data governance policies and compliance.
2. Identifying Purposes – Nata Supermarket must comprehensively define why the data is collected and needed.
3. Consent – Customers must provide informed and explicit consent before any data is used for data analysis or modelling tasks.
4. Limiting Collection – Only relevant data that is explicitly stated must be collected.
5. Limiting Use, Disclosure, and Retention – Data collected must only be used for its intended purpose and retained only as long as necessary.
6. Accuracy – Customer information must be accurate and updated.
7. Safeguards – Nata Supermarket must implement sufficient data protection measures.
8. Openness – Nata Supermarket must uphold transparency to customers in terms of data handling practices.
9. Individual Access – Customers have the right to view, correct, and request for deletion of data.
10. Challenging Compliance – Nata must have clear procedures outlined for customers to send complaints or concerns.

PIPEDA governs the collection, use, and disclosure of personal information during commercial activities of private-sector organisations in Canada, including federally-regulated businesses. Its objective is to safeguard personal information, while supporting electronic commerce. This is directly

applicable to Nata Supermarket since the company utilizes websites for product transactions In compliance with PIPEDA, Nata Supermarket should not share or sell consumer data without explicit consent from the user. These regulations imposed pushes Nata Supermarket to ensure transparency and fairness in their AI algorithms and data-driven decision-making processes. Most importantly, it develops legal responsibilities for Nata Supermarket utilizing customer data in business processes. Businesses that strictly adhere to Data Privacy Regulations (DPR) foster trust and credibility with their consumers, allowing them to engage more with their business.

Ethical Implications

Nata Supermarket's case study emphasizes this issue. The company collects extensive customer data to optimize inventory and promotions while overlooking the customer awareness and privacy risks. This may raise ethical responsibility and concerns in data collection practices if there is a lack of explicit customer consent. For example, TikTok's data collection practices have faced global scrutiny because of the increasing concerns regarding extensive data collection without adequate transparency. It was investigated that TikTok monitored users' personal information and behaviour, which led to consecutive bans in particular countries and regulatory fines. This accentuates the ethical significance of ensuring transparency and accountability of data collection processes across platforms.

Transparency, consent, fairness, and equity are ethical implications of using data and information in supporting business processes since they affect privacy, trust, and social responsibility (Viva Technology, 2025; Jayan, 2024). Starting off with transparency, it ensures that companies must clearly and openly communicate how they utilize personal data. When organisations fail to disclose proper data practices, ethical implications may arise and can result in potential breaches or misuse of trust. Businesses may risk their reputation, and users may feel deceived if there is no transparency to uphold. This emphasizes the importance of outlining privacy policies, data-sharing practices, and enabling users to manage their personal information. Consent is a vital ethical principle in data collection. This assures individuals are fully aware and have agreed to share their data and understand its intended use. Companies failing to explicitly obtain informed

consent are highly unethical since it violates an individual's privacy rights and personal autonomy. Businesses that prioritize user consent make it easier for individuals to withdraw permission at any time.

Fairness ensures that no community, group, or certain demographics are unfairly treated, excluded, or disadvantaged due to biased data practices. Ethical implications arise when businesses use biased and discriminatory datasets or algorithms. Equity ensures that all individuals have fair and equal access to resources and opportunities, regardless of socioeconomic status, gender, race, or group. Companies must continuously work on creating inclusive datasets that represent diverse populations and unbiased decision-making.

Data Threats

1. Malware – It is a malicious code or program that harms systems, such as viruses, worms, spyware, and Trojan horses (Gargiulo, 2022). These threats are designed to damage, disrupt, or gain unauthorized access to sensitive information by infecting devices through email attachments, files, or fraudulent downloads. These attacks frequently result in financial loss, identity theft, and compromise of the entire network infrastructure.
2. Phishing Attack – It is a type of online fraud where attackers or hackers attempt to steal confidential information such as credit card details, passwords, or bank account data by disguising themselves as a trustworthy entity or source (Rashid and Leyden, 2020). There are various types of phishing attacks, such as email phishing, spear phishing, whaling, smishing, vishing, and clone phishing.
3. Data Breaches – It is where unauthorized parties gain access to protected, sensitive, and confidential personal data or corporate data (Danao, 2023). Consequences of data breaches include severe financial losses, legal implications, reputational damage, and potential harm to victims. This generally happens when there are weak security measures, human errors, and the presence of hackers.

Mitigation Measures

1. Personal-Level Measures
    a. Password Management – It is important to use a complex and unique password in each account or use password managers (Numentis,

2023). Prevent predictable or generalized password patterns. Therefore, implementing a 12-character length with a combination of uppercase, lowercase, numbers, and symbols is ideal to prevent unauthorized access to accounts.

b. Authentication Protocols – It is essential to enable Two-Factor Authentication (2FA) since it is a two-step authentication to verify the user's identity, beyond usernames and passwords (Cisco, 2018). This adds an extra layer of security to user logins to protect against many security threats.
c. Personal Device Security – It is imperative to keep the operating systems of personal devices updated (Hetler, 2022). Along with that, installing reputable antivirus software and utilizing virtual private networks (VPNs) can significantly enhance device security. Avoiding or being cautious of public Wi-Fi networks, regularly backing up critical data, and enabling device encryption are also important protection methods for personal devices.

2. Organisation-Level Measures
    a. Employee Cybersecurity Training – There are emerging and evolving cyberthreats that require the workforce to be informed about in order to protect their personal data and reduce organisational risks (Luna, 2024). It is important to educate staff about the potential IT risks and vulnerabilities, which allows them to identify security threats by themselves in their workspace.
    b. Encryption of Sensitive Data – Transforming sensitive business information or data into an unreadable ciphertext, which prevents unauthorized access. This can be only decoded with a unique key. Businesses can significantly reduce the risk of data theft and also comply with industry regulations and government policies.
    c. Firewalls – They primarily regulate network traffic by blocking or permitting data packets based on predefined security rules (paloalto, 2015). Suspicious users are prevented from entering internal networks, while legitimate users are free to proceed. This maintains the confidentiality and integrity of sensitive data.
    d. Intrusion Detection Systems – It offers real-time tracking and analysis of network activity to spot potential threats such as unusual behaviour, known attack signatures, and policy violations, notifying

administrators for further investigation (paloalto, 2015). IDS and Firewall complement each other by safeguarding organisational assets and the network.

Implication of Data and Information to Nata Supermarket

Nata Supermarkets aims to remain competitive and foster business growth by utilizing data and information to support its business processes. This data-driven approach has multi-faceted implications for the company, which can impact customer satisfaction, operational efficiency, strategic planning, and competitive advantage.

1. Operational Efficiency
    - Efficient operations management can aid in improving revenue and reducing the costs of a supermarket. Supermarkets must incorporate a combination of strategies, including technology adoption, quality control, employee training, and lean management, to achieve operational efficiency (Anh, 2023). Nata Supermarket utilizes data and information to optimize operational efficiency through data-driven approaches, notably in demand forecasting and inventory management. Nata Supermarket can optimize stock levels based on thorough purchase patterns by precisely monitoring product spending on different categories, such as fruits, wines, meats, and sweets. This allows the supermarket to manage its inventory punctually and reduce waste. Furthermore, machine learning algorithms are capable of forecasting customer-segment-specific purchasing trends, seasonal fluctuations, and possible inventory surpluses or shortages. The company has a long-standing tradition of data-informed decision-making and utilization, maintaining ledgers with item locations, customer interactions, and detailed transaction records, which have assisted in less-expensive expansion and efficiency in stocking.
    - Kalli (2025) conducted a study on the Fast-Moving Consumer Goods (FMCG) sector wherein he highlights the critical role of Big Data Analytics in improving decision-making, optimizing supply chain, and providing timely operational insights to significantly enhance inventory, procurement, distribution, and logistics. Similarly, modern analytics tools such as Python, R, Excel, and Tableau can further

improve operations by enhancing demand predictions and promotion targeting. In addition, by aligning staffing levels with peak shopping hours in real-time, this significantly optimizes the workforce management. Comparable to how Walmart enhanced checkout optimization and pharmacy efficiency through predictive modelling, Nata Supermarket can improve its streamlining of processes, logistics, and sustain business growth by incorporating advanced data analytics into its overall business strategy.

2. Customer Satisfaction
    - Rosario and Diaz (2023) emphasize the increasing importance of data-driven marketing, wherein they stated that companies utilizing consumer and market data can enhance decision-making, gain a competitive edge, and strengthen company-customer relationships. Nata Supermarket leverages data and information to carry out sophisticated targeted marketing strategies and customer segmentation to improve customer personalization and experience.
    - The company can create precise customer profiles by analyzing demographic information, purchasing behaviours, household composition, and promotional responsiveness. This allows the deployment of personalized marketing campaigns, refining promotional strategies, and tracking acceptance rates. Therefore, this improves the effectiveness of marketing efforts in influencing customer behaviour. Moreover, customer service problems can be resolved through customer feedback analysis to enhance satisfaction. According to the study of Basariya and Mohammad (2021), key factors such as staff responsiveness, product quality, billing accuracy, and promotional offers greatly impact customer satisfaction in supermarkets.
    - Data-driven loyalty program and targeted promotions encourage more customers to continue buying the products or services of Nata Supermarket. This will increase repeated purchases and foster customer retention. Since the company has a strong customer-centric foundation to begin with, they have higher chances to stay competitive, especially by utilizing predictive analytics to tailor shopping experiences and optimize product stocking.
    - Similarly, Nata Supermarket can follow Walmart's approach, wherein they used purchase analysis and segmentation to inform product assortment decisions, boosting customer satisfaction and engagement. By constantly utilizing data-driven insights, Nata

Supermarket can strengthen and gain customer relationships and brand loyalty.

3. Competitive Advantage
    - Nata Supermarket utilizes data and information to maintain a competitive advantage by monitoring market trends, optimizing pricing strategies, and understanding customer behaviour. Competitive advantage is defined as the factors that enable a company to produce services or goods better or more cheaply than its competitors, which leads to higher margins or increased sales. There are different categories of competitive advantage.
    - Comparative advantage is where a firm produces products productively at a cheaper cost than their rivals, while differential advantage is where customers perceive products as superior due to factors like technology, reputation, branding, or customer service (Twin, 2024). Nata Supermarket can develop adaptive and innovative business strategies to counter market disruptions from technology-driven retailers and delivery services by benchmarking against competitors.
    - Nata Supermarket can introduce brand-new products that align with customer demand through data-driven insights, which ensures differentiation in a highly competitive retail environment. In addition, the ability to immediately respond to changing customer preferences helps create a unique value proposition. Several competitors in the retail industry already utilize data analytics to drive growth. In an article written by Ibanez (2021), it shows that retailers effectively utilizing data analytics can heighten profitability by up to 60%, since using customer data enables businesses to improve pricing strategies, personalize customer experience, and optimize inventory management.
    - In order for Nata Supermarket to be competitive, they can utilize an internal dataset for competitive intelligence, which maximizes profit margins, refines pricing strategies, and enhances market responsiveness (Twin, 2024). Economic moats or these sustainable advantages make it harder for competitors to replicate the strategic edge of Nata Supermarket, which ultimately boosts revenue and reinforces its market position.
4. Strategic Planning
    - Strategic Planning is a structured process that allows organisations to map out their long-term vision, set specific objectives, and lay out

strategies or steps to achieve them. It acts as a navigation system, which ensures that both short and long-term goals align with broader organisational objectives. This process grants numerous benefits, such as proactively setting goals and aligning teams toward a common objective, fostering a strong organisational culture, and ensuring that resources are allocated efficiently. In addition, businesses can assess opportunities and risks, enabling them to adapt to market conditions effectively.

- In the case study of Nata Supermarket, Vina Verago recognized the relevance and importance of utilizing analytics to improve strategic planning. By collecting and analyzing customer data, predictive modelling can generate insights to make informed decisions on market targeting, promotional strategies, and product stocking. This aligns with ClickUp's Supermarket Strategic Plan, wherein it highlights setting clear goals, tracking progress to measure success, improving customer experience, and outlining strategies for sales growth.
- Data and information play a vital role in long-term decision-making, assisting business expansion planning, mitigating risks, and identifying growth opportunities. Integrating structured strategic planning tools enables supermarkets to improve efficiency through KPIs, custom statuses, workload management, and collaboration features (ClickUp, 2025). This systematic approach ensures that supermarkets, like Nata Supermarket, can constantly enhance, stay competitive, and adjust to dynamic market conditions effectively.

Data and information play an important role in enhancing inventory management, optimizing supply chain logistics, personalizing marketing strategies, and improving overall business performance.

1. Inventory Management

   Inventory management heavily depends on real-time data collection, automation, and predictive analytics to ensure the alignment of stock levels on demand, while maintaining optimal inventory. According to the study of Karki (2024), there are specific ways in which data and information can be useful in inventory management:

   a. Automated Reordering Systems – To ensure timely replenishment of stock, AI-driven inventory management systems monitor stock levels and trigger automatic reordering built upon pre-set thresholds.
   b. Demand Forecasting – Predictive analytics utilizes historical sales data, seasonal trends, and external factors such as customer behaviour and

economic conditions in order to predict demand accurately. This assists businesses in avoiding overstocking and stock outs.

c. ABC Analysis or Classification & Grouping – Businesses categorize inventory based on consumption rate and value, which optimizes procurement strategies and the store. Goods that are high-value and fast-moving receive high priority in order planning.

2. Supply Chain Logistics

The incorporation of data, information, and advanced analytics into supply chain logistics has revolutionized traditional supply chain management into a more predictive, responsive, and efficient system. There are several ways in which artificial intelligence, cloud-based solutions, and machine learning are improving overall supply chain management (GeeksforGeeks, 2024; Wingfield, 2023; Thelosen, 2023):

a. AI-Optimized Planning and Forecasting – Google's Data and AI Cloud for Supply Chain offers AI-powered predictive tools that help businesses forecast demand and optimize inventory planning. Machine learning algorithms examine historical sales data, market fluctuations, and seasonal patterns to ensure accurate forecasting of demand.

b. Real-Time Supply Chain Visibility – Real-time tracking of goods is crucial for ensuring timely deliveries and managing disruptions. With the utilization of cloud platforms, advanced analytics, and Internet of Things (IoT) devices, organisations can acquire end-to-end visibility into the movement of raw materials, goods, and finished products. Overall, this enhances the coordination between distributors, manufacturers, and suppliers, improving relationships.

c. Route Optimization and Cost Reduction – Transportation costs profoundly affect overall supply chain efficiency. Organisations use AI-driven route optimization tools to analyze weather conditions, traffic patterns, and delivery schedules to choose the best routes. Therefore, this reduces transit times, fuel costs, and improves the reliability of delivery.

3. Marketing Strategies

Customers expect businesses to understand their preferences and tailor experiences. Data science has been transformative to the marketing aspect by allowing businesses to analyze big customer data, predict behaviors, and develop hyper-personalized campaigns. According to MITSDE (2024), there are many ways in which data and information improve marketing strategies:

a. Customer Segmentation – Customer segmentation enables businesses to group customers based on behavior, preferences, and demographics to create customer profiles.
b. Dynamic Content Personalization – Mobile apps, emails, and websites adjust content based on user behavior. Machine learning algorithms examine vast datasets to provide relevant content recommendations and products to users. This is done through content-based filtering and collaborative filtering in order to reflect individual interests and preferences.
c. Optimized Marketing Channels – Data analytics helps identify the most effective marketing channels to reach targeted audiences. To help brands allocate resources efficiently, insights from email response rates, website traffic, and social media engagement are useful.

Data Science in Nata Supermarket

Tools and Technologies in Data Science for Retail Operations

According to Condor (2025), POS in supermarkets automatically provides real-time updates and remote access to inventory levels, which prevents stockouts and overstocking while providing insights to make timely and informed decisions. Integration of POS systems allows retailers to have contactless payments like digital wallets and tap-to-pay solutions to suit customer preferences for safer and faster shopping transactions. It has incorporated customer data and loyalty programs, which help retailers to offer discounts and promotions to customers. POS systems provide valuable insights into sales trends and customer preferences. Beyond POS systems, Customer Relationship Management (CRM) Software helps the supermarket to stay ahead of the industry. Similarly, CRM is highly important for grocery stores since retailers must keep track of delivering a seamless shopping experience to retain customers, especially with the increasing competition from e-commerce giants (Walmart and Amazon) and online grocery stores Teknobgt (2025). Through customer data analysis, Nata Supermarket can personalize its product offerings to meet customer needs, build customer loyalty, increase sales, and streamline operations. This helps Nata Supermarket to create targeted marketing campaigns and personalized promotions to stay ahead of the highly competitive grocery industry. Lastly, the Internet of Things (IoT) has a revolutionary impact on grocery retail,

notably concentrating on inventory management systems and smart stores (Iseal, 2023). IoT has a lot to offer, such as smart shelves, automated checkout systems, and RFID sensors in order to greatly increase customer satisfaction, allow seamless shopping, and product recommendations. Additionally, IoT-powered inventory management technology enables predictive demand forecasting, real-time stock monitoring, automated reordering, and supply chain optimization to ensure delivery efficiency and product freshness.

In terms of data storage, Ramadan (2023) states that a large amount of data needs to be stored properly and organized so that it can be accessed easily and quickly when needed. This can be achieved through the creation of a database. It is the collection of related data, tables, or records stored in a particular storage medium. With the integration of Database Management Systems (DBMS) such as RDBMS and MySQL, Nata Supermarket can efficiently store and manage structured sales, customer data, and inventory for analytics. Apart from DBMS, Langdoc (2023) discussed how AWS solutions or cloud computing can help transform the grocery industry. One of the solutions includes AI/ML-based category management and inventory planning. AWS managed service called Amazon Forecast utilized AI and ML to assist supermarket owners to accurately and easily forecast business outcomes, which led to an improvement of 25 to 75 percent forecast accuracy, based on a real 600-store grocery chain in India. AWS provides real-time product recommendations to customers, which leads to greater offer conversion, market baskets, and more overall revenue. With AWS, Nata supermarket can build, train, and implement ML models for important situations with control over tools and workflows.

In line with predictive analytics and machine learning techniques, these are powerful tools for forecasting future sales volume for retailers like Nata Supermarket to make informed decisions (Mounika, 2022). These predictive models can be built using XGBoost, polynomial regression, random forest, and linear regression techniques. XGBoost is ideal for efficiency and scalability, making it a good choice for Nata Supermarket, which is aiming to optimize inventory and promotion strategies, especially its capability to analyze trends and influencing factors. In addition, time series forecasting for grocers helps navigate them through the complexities of

business operations (Naruka, 2024). For example, accurate predictions minimize product or food wastage through demand forecasting for perishable products, which ensures optimal inventory levels. Clustering algorithms will also aid Nata Supermarket to better understand customer purchasing behavior, especially with the rise of online retail platforms (Shobayo and Ogunleye, 2023). Customer segmentation is highly critical for enhancing customer-centric service, thus improving profitability. The study utilized different clustering techniques such as state-of-the-art (SOTA) clustering, K-means clustering, Gaussian mixture model (GMM), and DBSCAN. The study shows that GMM outperformed the other clustering methods, with the use of an online retail dataset.

Regarding data visualisation tools, Power BI and Tableau are popular and the top choices for turning data into visualisations. According to Tableau (2025), a supermarket chain, Whole Foods Market, transformed the company's everyday approach to customer experience and operational efficiency. Different types of dashboards can be created, such as (1) a promotional optimization dashboard to visualize the promotional cycle, so the retailer can prepare for availability issues. (2) product availability dashboard targets to solve the issue of having the right product at the right inventory levels, and lastly, (3) online shopping cart analysis helps retailers understand how marketing campaigns, customer behaviour, and other external factors affect shopping cart habits. Overall, retailers develop interactive dashboards or visualisations to create large business value.

Benefits of Using Data Science to Solve Problems in Nata Supermarket

- Improved inventory turnover – With data science, Nata Supermarket can utilize advanced analytics and machine learning algorithms to forecast which products or goods will sell faster to reduce overstocking or understocking issues. This can optimize stock replenishment schedules since it ensures that the goods are sold before they expire, which minimizes losses (CFI Team, 2025).
- Optimized pricing strategies – Data science helps retailers to identify the price elasticity of how the quantity supplied or demanded of the products changes when their price changes as well (Vipond, 2024). This allows grocery retailers to understand how various price points influence sales volume.

- Enhanced customer satisfaction – Personalized recommendations and targeted promotions make the shopping experience more convenient and enjoyable. For example, recommending complementary products like bread and cheese helps customers decide which products are suggested to be bought together. Data-driven decision-making enables grocers to develop promotions adjusted to specific customer groups.

Challenges and Limitations of Data Science Implementation

- Data Privacy and Security Concerns – Handling sensitive customer data such as personal information, purchasing behavior, and transactions necessitates stringent data security and privacy measures (GeeksforGeeks, 2024). This highlights the importance of compliance with data protection regulations like PIPEDA, GDPR, and CCPA. Robust security measures safeguard personal data from hacks and unauthorized access.
- Resource Constraints (Shalomova, 2023)
    - Data Constraints – There is a lack of high-quality or sufficient data to train models effectively. Also, certain data has legal or regulatory access restrictions.
    - Technological Constraints – Big data requires high-performance hardware and software to run and train models. Absence of scalable cloud-based solutions can hinder the implementation.
    - Manpower Constraints – There is a need for skilled data scientists or data experts to fully implement and utilize this advanced technology.
    - Budgetary Constraints – Data science involves training, licensing, and project costs.

Proposed Solution

Design Phase

The design phase serves as the strategic foundation for Nata Supermarket to align data science methodologies to the business goals. This phase understands the operational needs of the organisation, identifying the most suitable data-driven solutions, and planning the implementation of different techniques to ensure the derived insights are meaningful and actionable. To start the design phase, a thorough review of the dataset and case study was done. Nata Supermarket, a retailer based in Brendan, Manitoba, Canada, pursue to maximize the utilization of customer and transactional data. The organisational goals identified were:

- Optimizing targeted promotions to increase ROI and campaign acceptance.
- Enhancing product inventory management through demand forecasting and identification of high value products.
- Gain valuable insights into customer behaviour and preferences through segmentation and trend analysis.
- Maintaining a competitive edge in the retail industry by making data-driven decisions.

The design of the program address Nata Supermarket's needs as follows:

1. Customer Behaviour Analysis through Exploratory Data Analysis
    - EDA includes techniques such as statistical summaries, data visualisations (bar plot, histograms, heatmaps), and correlation analysis. This is done to identify patterns, trends, spending patterns, product preferences and get a grasp on their customer base. With that, it provides insights into customer preferences that assist pricing strategies, product bundling, and inventory management.
2. Customer Segmentation using Clustering
    - K-means clustering is used to group customers based on demographic data and purchase patterns, which enables targeted marketing strategies. Principal Component Analysis (PCA) is applied to simplify complex multivariate data and emphasize the influential features, making the patterns more interpretable. This supports better decision-making in product strategy and campaign. Furthermore, by distinguishing customer segments (e.g., high spenders vs budget spenders), Nata Supermarket can adjust promotions to boost customer engagement and retention.
3. Predictive Modelling for Campaign Acceptance
    - Classification models such as logistic regression, random forest, decision tree, and XGBoost are trained to classify whether a customer is likely to accept a marketing campaign. This predicts campaign success, allowing Nata Supermarket to allocate resources more effectively and tailor targeted promotions, which is one of the main challenges of Nata Supermarket.
4. Sales Forecasting and Time Series Analysis

- Augmented Dickey-Fuller (ADF) Test, ARIMA, and SARIMAX models are implemented to forecast future sales trends for main product categories over time. This significantly helps with inventory planning and ensure that the supermarket can meet demand fluctuations throughout the years.

The tools and techniques were selected based on their suitability for the problem context and the Nata Supermarket dataset:

1. Python with libraries like pandas, numpy, xgboost, scikit-learn, statsmodel, and plotly for effect data processing, transformation, modelling, forecasting, and visualisation.
2. Data preprocessing methods including outlier detection, missing value imputation, and feature scaling are done to ensure model accuracy and data quality.
3. To validate classification models, evaluation metrics such as accuracy, precision, recall, F-1 score, and AUC-ROC are used.
4. Hyperparameter tuning and cross-validation were implemented for model generalizability and optimization

Each design choice aligns with Nata Supermarket's needs and objectives. This ensures that the final implementations not only offers a technically sound solution but also converts into something valuable and actionable for Nata Supermarket.

Goal	**Method**	**Result**
Optimize Targeted Promotions	Clustering, Predictive Modelling, EDA	Tailored campaign or promotion strategies.
Improve Inventory Management	Time Series Forecasting, EDA	Forecast product demand trends.
Understand Customer behaviour	EDA, PCA	Identified customer profile preferences and behaviours.
Stay Competitive	Data Visualisation and Insights	Support strategic planning.

Implementation Phase

Data Preprocessing and Cleaning

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from datetime import datetime
from sklearn.cluster import KMeans
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.decomposition import PCA
from sklearn.metrics import silhouette_score
from kneed import KneeLocator
from sklearn.metrics import silhouette_samples
import matplotlib.cm as cm
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.tree import DecisionTreeClassifier
from xgboost import XGBClassifier
from sklearn.cluster import AgglomerativeClustering
import scipy.cluster.hierarchy as sch
from sklearn.metrics import confusion_matrix, accuracy_score, roc_auc_score, classification_report
from imblearn.over_sampling import SMOTE
from imblearn.pipeline import Pipeline
from statsmodels.tsa.arima.model import ARIMA
from statsmodels.tsa.statespace.sarimax import SARIMAX
from statsmodels.tools.eval_measures import rmse
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from statsmodels.tsa.stattools import adfuller
import plotly.graph_objs as go
```

```
df = pd.read_csv("C:\\Users\\amand\\Downloads\\YEAR 2\\Term 3\\Business Process Support\\nata.csv")
```

- The first imperative step in every data project is to import essential libraries involving manipulation, exploration, forecasting, and model evaluation. Numpy and Pandas are utilized for data manipulation and analysis, while Seaborn, Matplotlib, and Plotly are popular tools for creating visualisations. K-Means and PCA are used for clustering and dimensionality reduction, respectively. Logistic Regression, Decision Tree Classifier, XGB Classifier, and Random Forest Classifier are used to build predictive classification models. Sklearn.metrics tools are imported to assess model performance, while SMOTE and Pipeline resolve class imbalance problems and streamline model workflows. In addition, statsmodel library provides a set of tools for time series analysis using SARIMAX and ARIMA, and statistical methods like the Augmented Dickey-Fuller test for stationary. The Nata Supermarket's dataset is then read by the program and assigned to a variable.

```
[8]: df.head()
```

[8]:

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Customer	Recency	MntWines	...	NumWebVisitsMonth	AcceptedCmp3	AcceptedCm
0	5524	1957	Graduation	Single	58138.0	0	0	09/04/2012	58	635	...	7	0	
1	2174	1954	Graduation	Single	46344.0	1	1	03/08/2014	38	11	...	5	0	
2	4141	1965	Graduation	Together	71613.0	0	0	21-08-2013	26	426	...	4	0	
3	6182	1984	Graduation	Together	26646.0	1	0	02/10/2014	26	11	...	6	0	
4	5324	1981	PhD	Married	58293.0	1	0	19-01-2014	94	173	...	5	0	

5 rows × 29 columns

```
[9]: df.shape
```

```
[9]: (2240, 29)
```

```
[10]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2240 entries, 0 to 2239
Data columns (total 29 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   ID                   2240 non-null   int64
 1   Year_Birth           2240 non-null   int64
 2   Education            2240 non-null   object
 3   Marital_Status       2240 non-null   object
 4   Income               2216 non-null   float64
 5   Kidhome              2240 non-null   int64
 6   Teenhome             2240 non-null   int64
 7   Dt_Customer          2240 non-null   object
 8   Recency              2240 non-null   int64
 9   MntWines             2240 non-null   int64
 10  MntFruits            2240 non-null   int64
 11  MntMeatProducts      2240 non-null   int64
 12  MntFishProducts      2240 non-null   int64
 13  MntSweetProducts     2240 non-null   int64
 14  MntGoldProds         2240 non-null   int64
 15  NumDealsPurchases    2240 non-null   int64
 16  NumWebPurchases      2240 non-null   int64
 17  NumCatalogPurchases  2240 non-null   int64
 18  NumStorePurchases    2240 non-null   int64
 19  NumWebVisitsMonth    2240 non-null   int64
 20  AcceptedCmp3         2240 non-null   int64
 21  AcceptedCmp4         2240 non-null   int64
 22  AcceptedCmp5         2240 non-null   int64
 23  AcceptedCmp1         2240 non-null   int64
 24  AcceptedCmp2         2240 non-null   int64
 25  Complain             2240 non-null   int64
 26  Z_CostContact        2240 non-null   int64
 27  Z_Revenue            2240 non-null   int64
 28  Response             2240 non-null   int64
dtypes: float64(1), int64(25), object(3)
memory usage: 507.6+ KB
```

- The following code blocks quickly provide necessary information about the structure and contents of the Nata Supermarket's dataset. The .head() method displays the first five rows, the .shape() attribute indicates the size of the dataset—in this case, 2,240 rows and 29 columns, the .info() method provides a summary of the DataFrame, including the columns' data type and missing values. It can be observed that the Income column has 24 null values, and the dataset contains a combination of numerical and categorical data types. This preliminary information is useful for detecting data quality

issues and preparing essential preprocessing steps, including standardizing data types and handling missing values.

```
df.drop(columns=["ID", "Z_CostContact", "Complain", "Z_Revenue", "Teenhome", "Kidhome", "Response"], inplace=True)
```

```
df["Dt_Customer"] = df["Dt_Customer"].astype(str).str.replace("-", "/", regex=True)
df["Dt_Customer"] = pd.to_datetime(df["Dt_Customer"], format="%d/%m/%Y", dayfirst=True, errors="coerce")
```

```
df["Income"] = df["Income"].fillna(df["Income"].median())
```

- The first line of code removes unnecessary columns from the dataset that are irrelevant to the analysis. ID is a unique numeric identifier, while Z_CostContact, Z_Revenue, and Response lack context about the column and their values, making them best to exclude due to uncertainty. The Complain, Teenhome, and Kidhome columns are excluded as well, since they are not relevant to the modelling goal. The second and third lines of code ensure a consistent date format by replacing hyphens with slashes in order for the program to properly interpret the date string and prevent parsing errors. To handle the missing values in the Income column, those values are replaced with the median, which is a common imputation technique that reduces the influence of outliers.

```
df.isnull().sum()
```

```
Year_Birth            0
Education             0
Marital_Status        0
Income                0
Dt_Customer           0
Recency               0
MntWines              0
MntFruits             0
MntMeatProducts       0
MntFishProducts       0
MntSweetProducts      0
MntGoldProds          0
NumDealsPurchases     0
NumWebPurchases       0
NumCatalogPurchases   0
NumStorePurchases     0
NumWebVisitsMonth     0
AcceptedCmp3          0
AcceptedCmp4          0
AcceptedCmp5          0
AcceptedCmp1          0
AcceptedCmp2          0
dtype: int64
```

```
current_year = datetime.now().year
df["Age"] = current_year - df["Year_Birth"]
```

```
df.duplicated().sum()
```

```
201
```

```
df.drop_duplicates(inplace=True)
```

- After executing the imputation technique on Income, it was confirmed that the missing values had been resolved. Next, a new column called Age is created by subtracting the customer's birth year from the current year. This is done because Age is much practical and interpretable feature than the year of birth. The following step is to check the number of duplicate rows in the dataset. The result shows that there are 201 duplicates, which are all dropped to ensure that the dataset is consistent, clean, and reliable, as duplicates could potentially skew the analysis and lead to biased results. These steps are critical for effective data cleaning.

```
Q1 = df["Income"].quantile(0.25)
Q3 = df["Income"].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
df = df[(df["Income"] >= lower_bound) & (df["Income"] <= upper_bound)]
```

```
cols = df.columns.tolist()
age_index = cols.index('Age')
year_birth_index = cols.index('Year_Birth')

cols.remove('Age')

cols.insert(year_birth_index + 1, 'Age')

df = df[cols]

print(df.columns)
```

```
Index(['Year_Birth', 'Age', 'Education', 'Marital_Status', 'Income',
       'Dt_Customer', 'Recency', 'MntWines', 'MntFruits', 'MntMeatProducts',
       'MntFishProducts', 'MntSweetProducts', 'MntGoldProds',
       'NumDealsPurchases', 'NumWebPurchases', 'NumCatalogPurchases',
       'NumStorePurchases', 'NumWebVisitsMonth', 'AcceptedCmp3',
       'AcceptedCmp4', 'AcceptedCmp5', 'AcceptedCmp1', 'AcceptedCmp2'],
      dtype='object')
```

- The first code block is executed to detect and remove outliers in the Income column using the IQR or Interquartile Range technique. IQR measures the spread of the middle 50% of Income values. Values that are out of range, 1.5 times below Q1 and Q3, are considered outliers. In this way, noise is reduced and prevents extreme outliers from skewing the analysis by filtering rows within certain bounds.

Exploratory Data Analysis (EDA)

```
df.describe(include='all').T
```

	count	unique	top	freq	mean	min	25%	50%	75%	max	std
Year_Birth	2031.0	NaN	NaN	NaN	1968.753323	1893.0	1959.0	1970.0	1977.0	1996.0	11.989926
Age	2031.0	NaN	NaN	NaN	56.246677	29.0	48.0	55.0	66.0	132.0	11.989926
Education	2031	5	Graduation	1022	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Marital_Status	2031	8	Married	785	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Income	2031.0	NaN	NaN	NaN	51683.792713	1730.0	35686.0	51381.5	68117.5	113734.0	20599.686377
Dt_Customer	2031	NaN	NaN	NaN	2013-07-13 23:15:19.940915968	2012-01-08 00:00:00	2013-01-21 12:00:00	2013-07-14 00:00:00	2014-01-01 12:00:00	2014-12-06 00:00:00	NaN
Recency	2031.0	NaN	NaN	NaN	49.098474	0.0	24.0	49.0	74.0	99.0	28.964054
MntWines	2031.0	NaN	NaN	NaN	304.422452	0.0	23.5	177.0	504.5	1493.0	336.028618
MntFruits	2031.0	NaN	NaN	NaN	26.437223	0.0	2.0	8.0	33.0	199.0	39.876903
MntMeatProducts	2031.0	NaN	NaN	NaN	165.966519	0.0	16.0	68.0	229.0	1725.0	221.065915
MntFishProducts	2031.0	NaN	NaN	NaN	37.587395	0.0	3.0	12.0	50.0	259.0	54.771753
MntSweetProducts	2031.0	NaN	NaN	NaN	27.31807	0.0	1.0	8.0	34.0	263.0	41.751803
MntGoldProds	2031.0	NaN	NaN	NaN	43.917774	0.0	9.0	24.0	56.0	362.0	52.065747
NumDealsPurchases	2031.0	NaN	NaN	NaN	2.321516	0.0	1.0	2.0	3.0	15.0	1.892824
NumWebPurchases	2031.0	NaN	NaN	NaN	4.112752	0.0	2.0	4.0	6.0	27.0	2.795596
NumCatalogPurchases	2031.0	NaN	NaN	NaN	2.630231	0.0	0.0	2.0	4.0	28.0	2.797017
NumStorePurchases	2031.0	NaN	NaN	NaN	5.792221	0.0	3.0	5.0	8.0	13.0	3.227218
NumWebVisitsMonth	2031.0	NaN	NaN	NaN	5.326933	0.0	3.0	6.0	7.0	20.0	2.426142
AcceptedCmp3	2031.0	NaN	NaN	NaN	0.072871	0.0	0.0	0.0	0.0	1.0	0.259988
AcceptedCmp4	2031.0	NaN	NaN	NaN	0.076317	0.0	0.0	0.0	0.0	1.0	0.26557
AcceptedCmp5	2031.0	NaN	NaN	NaN	0.072378	0.0	0.0	0.0	0.0	1.0	0.259177
AcceptedCmp1	2031.0	NaN	NaN	NaN	0.065977	0.0	0.0	0.0	0.0	1.0	0.248304
AcceptedCmp2	2031.0	NaN	NaN	NaN	0.012802	0.0	0.0	0.0	0.0	1.0	0.112445

- The .describe() method provides the summary statistics of each column. The Age varies widely, with a mean of 56 years old. However, it is noticeable that 132 is the maximum age, indicating that there is a presence of data errors or outliers. On the other hand, there is a significant diversity among customers in terms of income, wherein the average is approximately 51,684. On marital status, most customers are graduates and are married, based on frequency counts. For products, wine is the most purchased product in the supermarket, followed by meat. Lastly, customers are more likely to visit and purchase in stores than via catalogue or websites.

```
num_cols = df.select_dtypes(include=['int64', 'float64']).columns
cat_cols = df.select_dtypes(include=['object']).columns
```

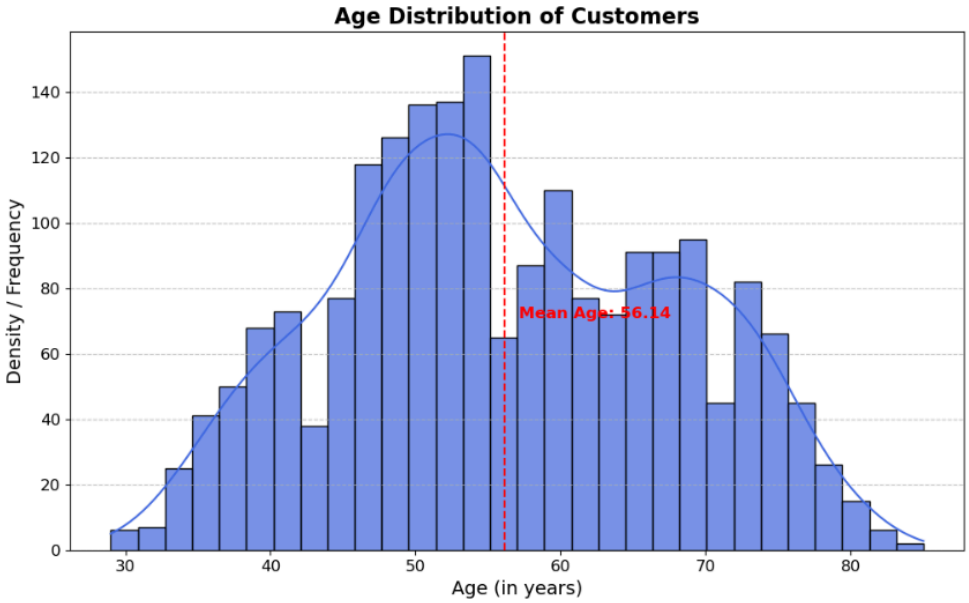
```
for col in cat_cols:
    print(f"\nValue Counts for {col}:")
    print(df[col].value_counts().head(10))
```

```
Value Counts for Education:
Education
Graduation    1022
PhD            438
Master         336
2n Cycle       186
Basic           49
Name: count, dtype: int64

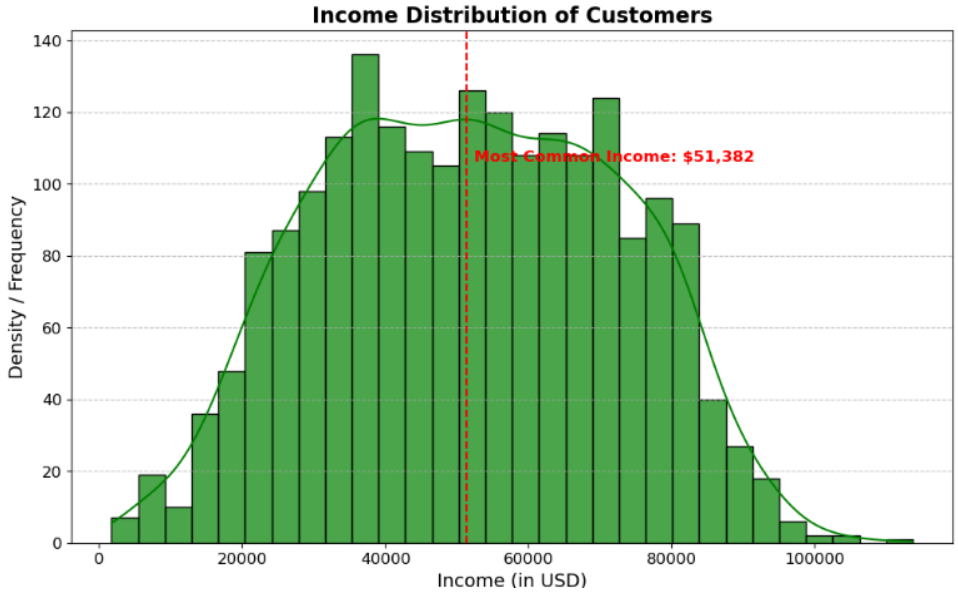
Value Counts for Marital_Status:
Marital_Status
Married     785
Together    513
Single      445
Divorced    212
Widow        70
Alone         3
Absurd        2
YOLO          1
Name: count, dtype: int64
```

- The first code block selects columns with numerical and categorical data types and stores them in the num_cols and cat_cols columns, respectively. The second code block counts and displays the frequency of the values in categorical columns, Education and Marital Status. This confirms that the majority of the customers have Graduation as the level of education, with 1022 occurrences. Meanwhile, Married (785), Together (513), and Single (445) are the top 3 most prevalent marital statuses.

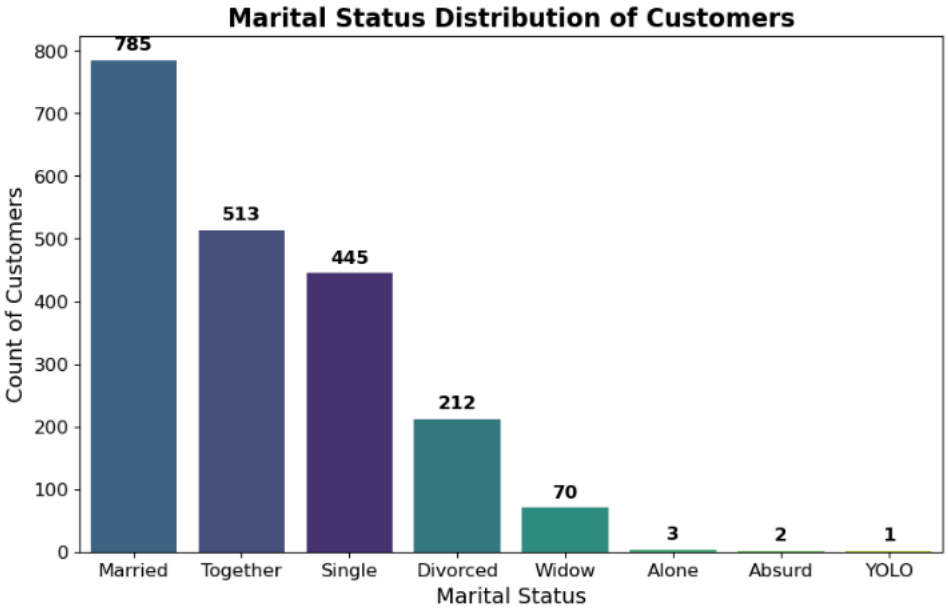
Demographics



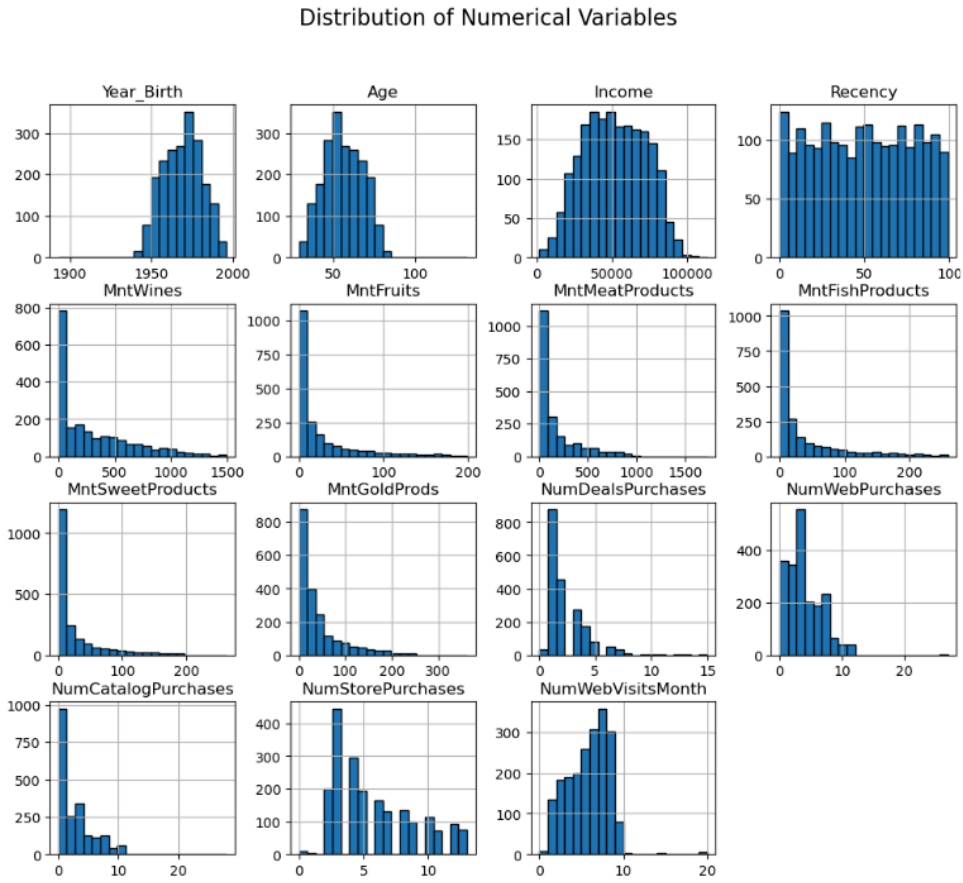
- The histogram shows the age distribution of customers at Nata Supermarket. Prior to this, outliers above the age of 100 were excluded, as they contribute to a slight negative skew. It is observed that the nature of the distribution is bimodal, which means that there are two distinct peaks in the data. The mean age is 56 years, as illustrated by the vertical dashed line. The primary peak occurs around ages 50–55 years old, while the secondary peak is around ages 65–70 years old. It can be concluded that the overall age range spans from approximately 30 to 85 years old, excluding the outliers.



- The second histogram shows the income distribution of customers at Nata Supermarket. It is identified that the most common income is $51,382. The distribution ranges from approximately $0 to beyond $100,000, with fewer customers at the extreme outliers. It is observed that the distribution appears to be partly symmetrical around the most common income, with a bit of positive skew. The highest density occurs around $35,000-$40,000, where the highest peak or bar reaches around 135 customers. On top of that, the spread of customers across the middle-income range ($30,000-$80,000) is fairly even, wherein there are multiple small peaks rather than a perfect bell curve.

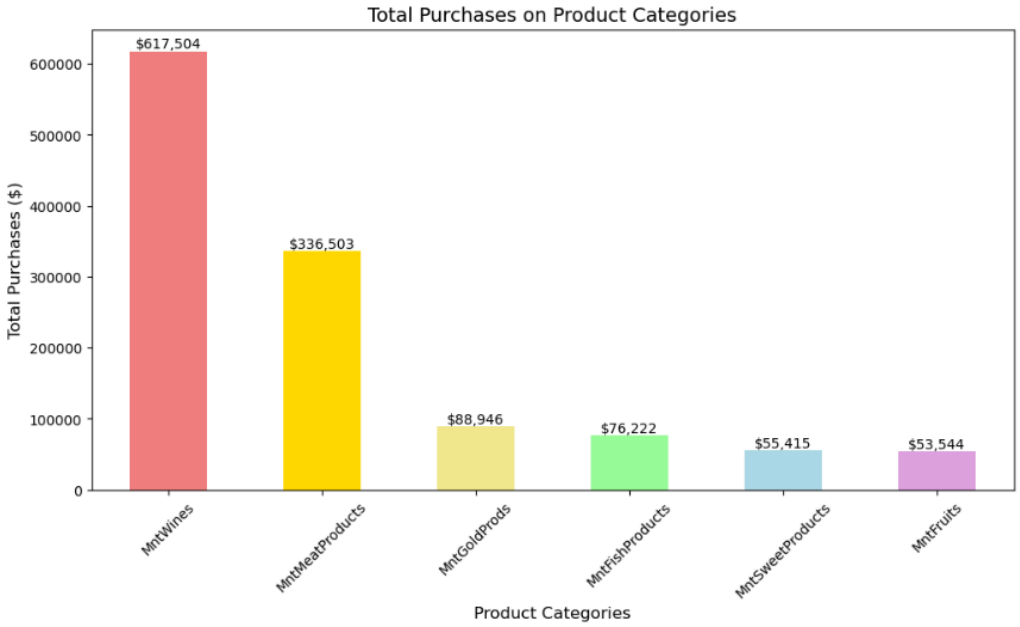


- The bar graph shows the distribution of marital status among customers at Nata Supermarket. It is evident that married customers dominate the distribution, making it the primary customer base, followed by smaller proportions of those who are together (presumably cohabiting), single, divorced, and widowed. There is a presence of unusual values or outliers, which are alone, absurd, and YOLO marital statuses.

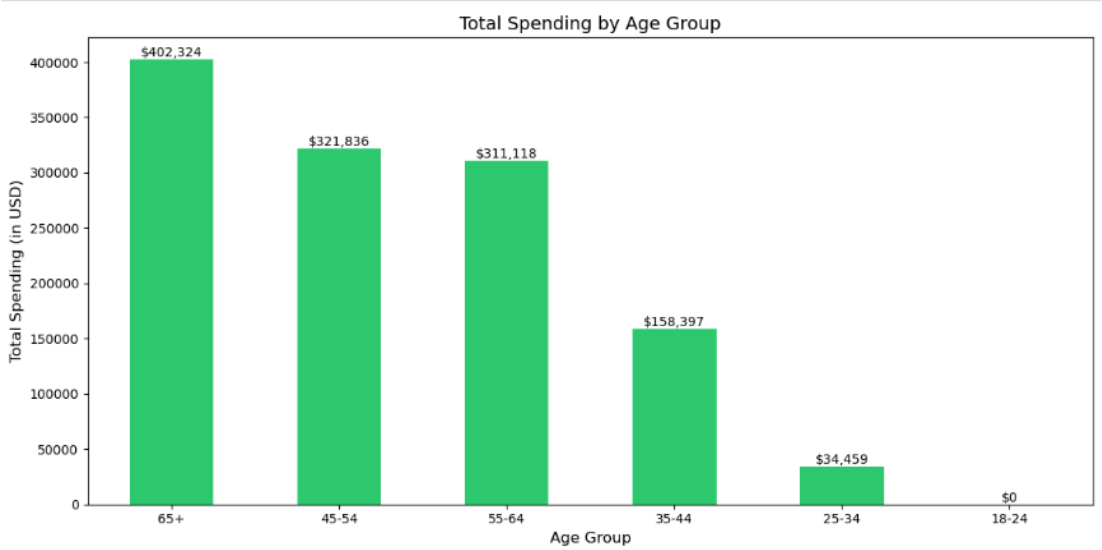


- The illustration shows the distribution of graphs for numerical variables in the dataset. Starting with the year of birth and age, it confirms the earlier analysis that most customers were born between 1950-1980, which corresponds to ages mainly between 35–75 years old. Recency is observed to have a fairly uniform distribution from 0 to 100 days, which implies consistent customer activity over this period. In terms of spending patterns, it is evident that there is a strong positive skew in wine purchases. The majority of customers spend small amounts (under $250), but with a long tail of higher spenders. Fruits, meat, fish, sweets, and gold products all show extremely right-skewed distributions, wherein most customers spend very little on each product category, with frequencies decreasing significantly as the amount spent increases. Similarly, purchase behaviour displays a right-skewed distribution, except for the number of in-store purchases, which appears to have an even distribution. Most customers make a few discounted purchases, 2–4 purchases on the web, 0–2 catalogue purchases, 1–10 in-store purchases, and 5-8 website visits per month.

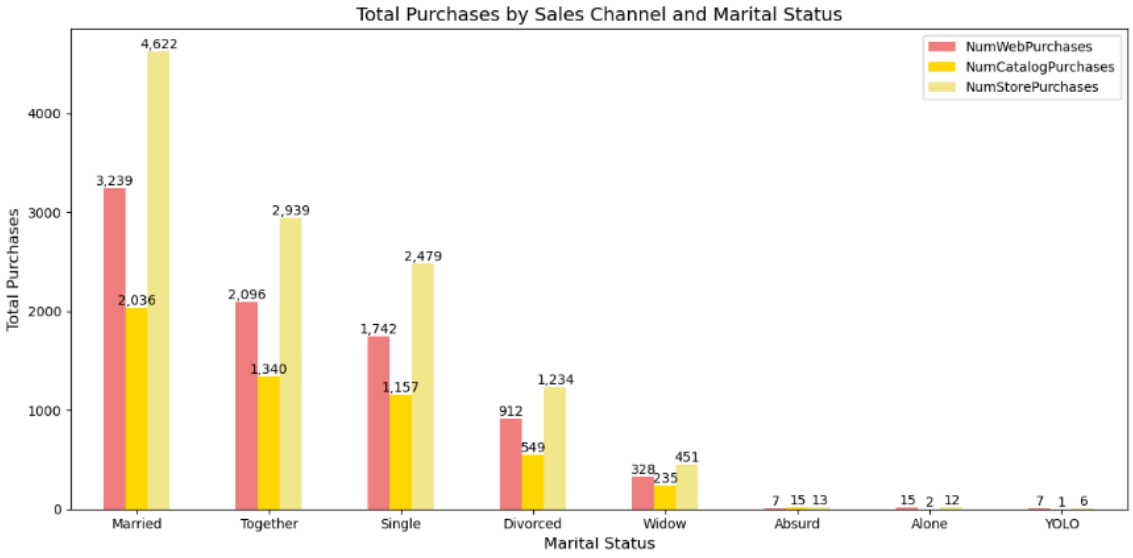
Spending Behavior



- Wines have the highest total purchase amount, amounting to $617,504, far surpassing all other product categories. This means that wine is the most popular and widely bought product category. Meat products come second with a $336,503 contribution. While it's less significant than wine, it still chips in a major share of the revenue of Nata Supermarket. The rest of the categories, such as gold, fish, sweets, and fruit products, show much lower total spending. It may imply that these products are purchased less frequently, priced lower per unit, or both.



- To obtain meaningful insights into customer spending across different ages, the data is segmented into different age groups or bins. Customers ages 65 and above are the highest-spending demographic, which contributes $402,324 in total spending, despite this age group not being the largest segment in the Nata Supermarket's customer base. Middle-aged customers aged 45-54 and 55-64 rank second in spending, with closely matched total expenditure levels of $321,836 and $311,118, respectively. There's a noticeable drop in spending for the 35-44 segment ($158,397), which is less than half of the amount spent by middle-aged groups. Meanwhile, young adults (25-34) spend minimally, with a total of $34,459 total spending, while the youngest age group (18-24) indicates no spending at all.

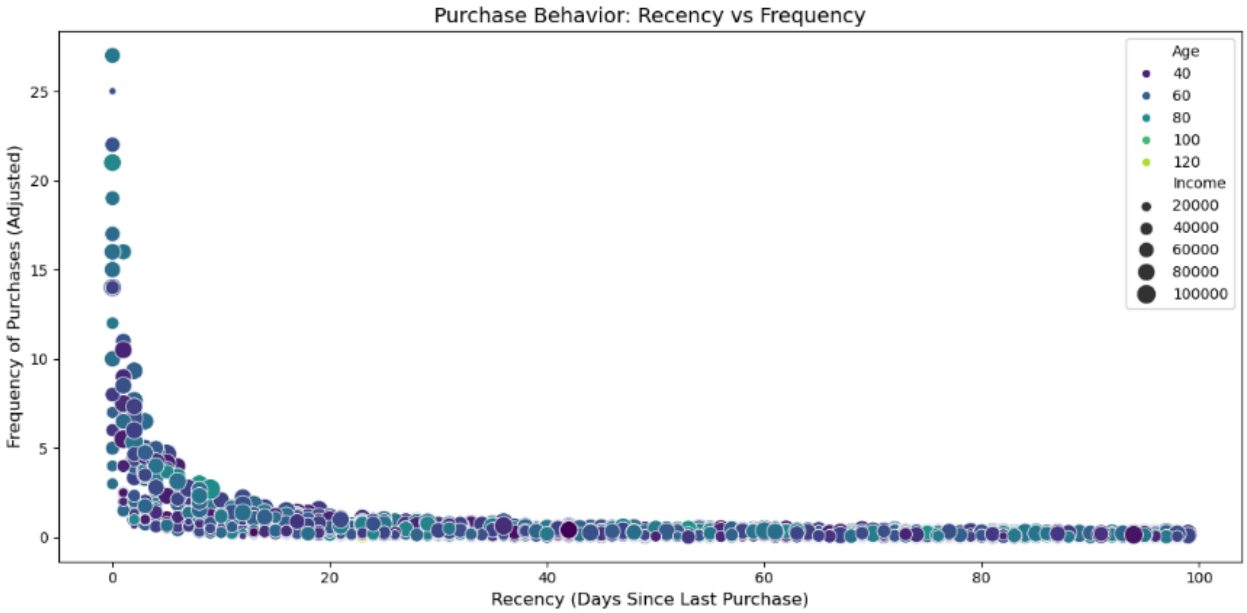


	Purchase Channel	Total Purchases
0	NumStorePurchases	11756
1	NumWebPurchases	8346
2	NumCatalogPurchases	5335

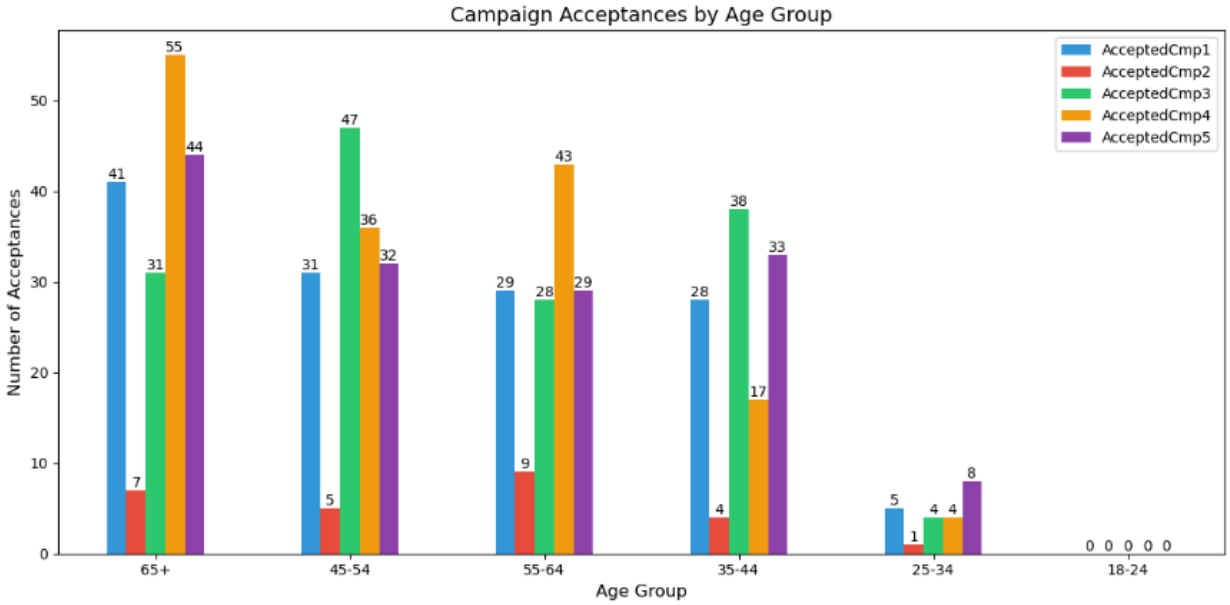
- In terms of purchase channel, in-store purchases dominate the dataset with a total of 11,756 transactions, while web purchases follow with 8,346 transactions. Catalogue purchases are the least frequent, amounting to 5,335. On the other hand, married customers are confirmed to be the highest purchasers across all channels, followed by together and single statuses. Divorced and Window categories indicate moderate activity, while absurd, alone, and YOLO groups have very minimal purchases.



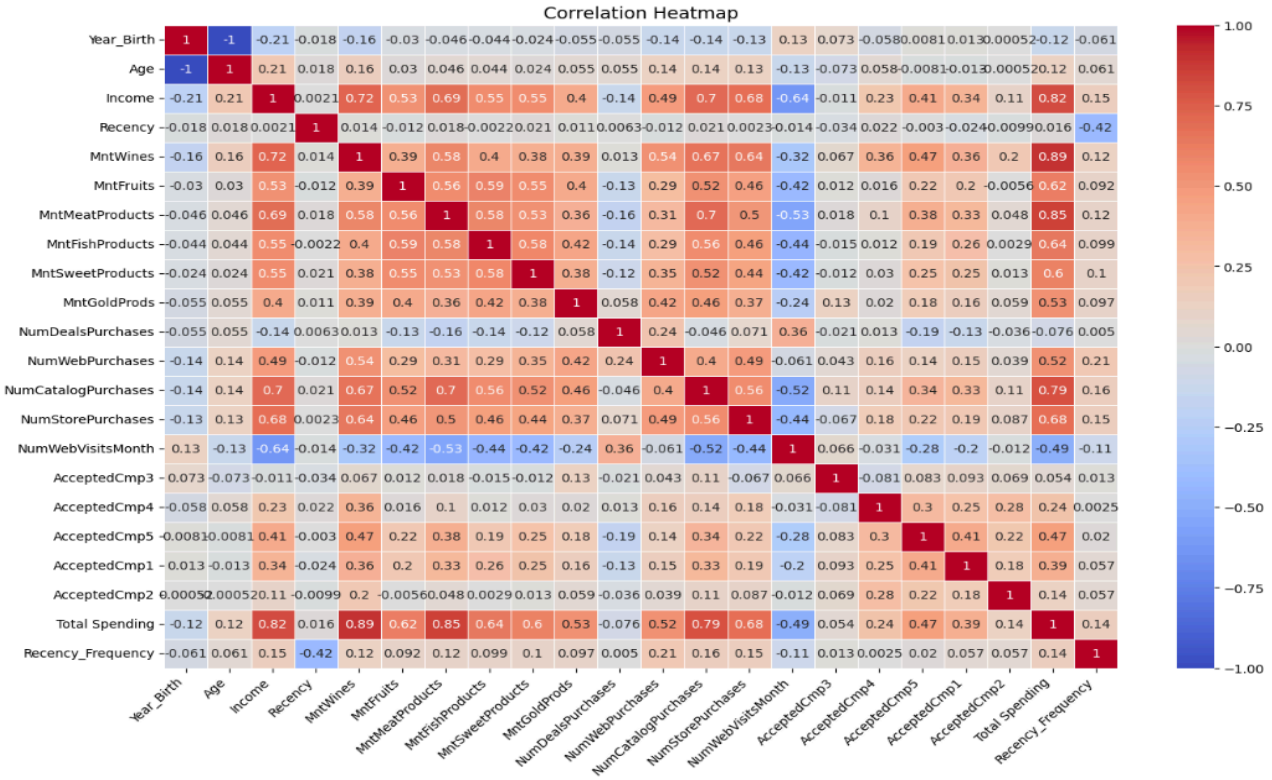
- The scatter plot shows the relationship between the number of online purchases and the number of website visits per month of Nata Supermarket customers. The data points are similar to a grid-like pattern, between 0-10 website visits and 0-11 online purchases, which indicates that this is where most customer behaviour falls. It appears that there is no strong linear correlation between website visits and online purchases, since there is no upward trend line that would imply more visits consistently lead to more customer purchases. A few evident outliers are shown in the upper left area, with customers making around 23-27 online purchases despite very low website visits (0-1 per month). This means that some shoppers are highly efficient or, most likely, make bulk purchases. Some customers visit several times (12 visits) but make little to no purchases, as evidenced by the concentration of data points across the bottom of the graph.



- The scatter plot illustrates the relationship between recency and frequency of customers, integrating additional dimensions of age (represented by colour) and income (represented by size of data point). The results show a significant negative correlation between recency and frequency. Simply put, customers who have purchased recently are more likely to shop more frequently overall, while those who haven't shopped in several days or months tend to exhibit lower frequency values. Most data points fall in the 0-20 days recency range, with a long tail extending to 100 days. The distribution appears to have a hyperbolic pattern, wherein the frequency declines drastically as recency increases. It is reasonable to state that customers or data points located on the left side represent frequent and active shoppers, the middle area corresponds to occasional shoppers, and the right area suggests lapsed customers. The colour coding of the data points reveals a mix of age groups, with certain ages like 40 and 60 years old dominating and spread across regions. In addition, there are 80-year-old data points that are determined as frequent shoppers. This diversity suggests that age alone isn't strongly deterministic of the relationship between recency and frequency. Lastly, larger dots or higher income customers are evident across the plot, but are especially visible among both the most frequent shoppers and the recent shoppers. To some degree, income may influence purchase frequency.



- The bar chart illustrates campaign acceptance rates across different age groups of customers at Nata Supermarket in order to find out patterns in marketing effectiveness. The 65+ age group, or older customers, respond best to campaign 5, middle-aged customers (45-54) respond best to campaign 3, and campaign 4 portrays consistent performance across different age groups. It is observed that campaign effectiveness clearly differs by age, wherein order demographics show significantly higher campaign engagement rates. It is important to point out that campaign 2 consistently underperformed on all age groups, which means there might be issues with its approach or content. The complete zero response to all campaigns from the 18-24 age group verifies earlier analysis that this specific demographic is not part of Nata Supermarkets' customer base.

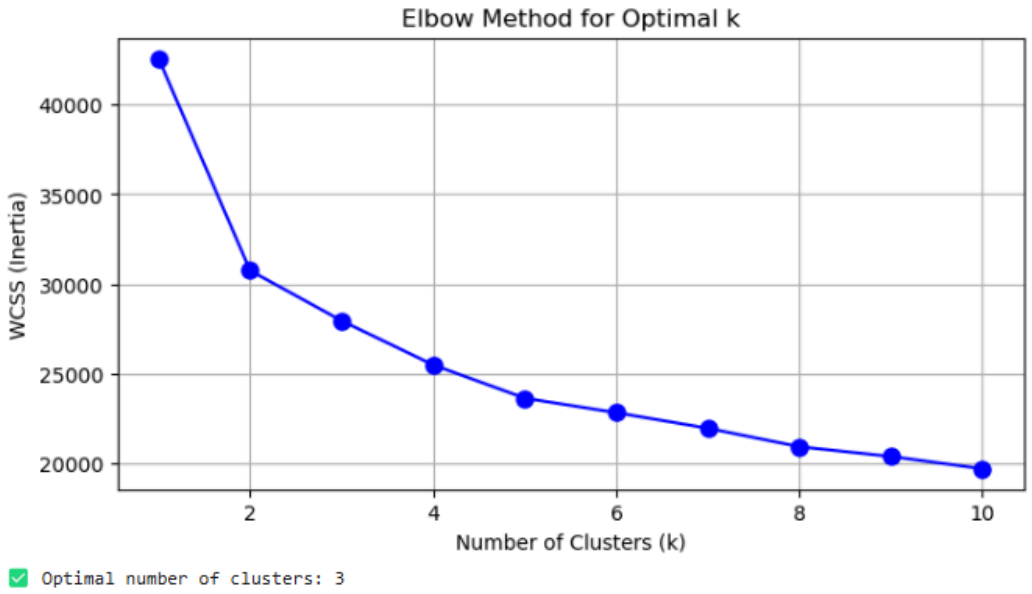



```
Relationships with correlation > 0.65:
Correlation between Age and Year_Birth: -1.00
Correlation between Year_Birth and Age: -1.00
Correlation between MntWines and Income: 0.72
Correlation between MntMeatProducts and Income: 0.69
Correlation between NumCatalogPurchases and Income: 0.70
Correlation between NumStorePurchases and Income: 0.68
Correlation between Total Spending and Income: 0.82
Correlation between Income and MntWines: 0.72
Correlation between NumCatalogPurchases and MntWines: 0.67
Correlation between Total Spending and MntWines: 0.89
Correlation between Income and MntMeatProducts: 0.69
Correlation between NumCatalogPurchases and MntMeatProducts: 0.70
Correlation between Total Spending and MntMeatProducts: 0.85
Correlation between Income and NumCatalogPurchases: 0.70
Correlation between MntWines and NumCatalogPurchases: 0.67
Correlation between MntMeatProducts and NumCatalogPurchases: 0.70
Correlation between Total Spending and NumCatalogPurchases: 0.79
Correlation between Income and NumStorePurchases: 0.68
Correlation between Total Spending and NumStorePurchases: 0.68
Correlation between Income and Total Spending: 0.82
Correlation between MntWines and Total Spending: 0.89
Correlation between MntMeatProducts and Total Spending: 0.85
Correlation between NumCatalogPurchases and Total Spending: 0.79
Correlation between NumStorePurchases and Total Spending: 0.68
```

- The correlation heatmap reveals important relationship patterns between the dataset variables. Starting with positive relationships, there is a strong correlation between wines, meat, and total spending. These relationships imply that wine and meat shoppers are the largest contributors to the overall revenue or spending of Nata Supermarket. In terms of income and spending behaviour, there is a notable correlation between income and total spending, wines, meat, catalogue purchases, and store purchases. These relationships with income confirm that higher-income consumers spend more in general,

especially on premium categories such as wine and meat. Lastly, the strong correlation between catalogue purchases and total spending, wines, and meat products suggests that catalogue shopping is connected with higher-value purchases, especially for wine and meat goods. On the other hand, it is important to note some negative correlations. Age and year of birth are in a perfect negative correlation, since they reflect the inverse mathematical relationship. Recency and Recency Frequency (-0.42) imply that customers who haven't purchased recently tend to have little purchase frequency in general. There is a negative correlation as well between web visits and product purchases (-0.42, -0.53, -0.44, and -0.42), wherein it's reasonable to state that browsing behaviour doesn't translate well to guaranteed purchases, which echoes earlier analysis of the effectiveness of the website. Regarding the marketing campaign, Campaign 5 demonstrates a moderate positive correlation between income, wine purchases, and total spending. This means that campaign 5 works well with older, affluent segments because it appeals to high-value customers. Campaign 3 indicates little correlation with purchasing behaviour, despite being prevalent with the 45-54 age group.

Clustering





Cluster Summary (Mean values per feature):

Cluster	Age	Income	Recency	MntWines	MntMeatProducts	MntFruits	MntFishProducts	MntSweetProducts	MntGoldProds	NumDealsPurchases
0	55.84	80139.19	50.37	840.17	474.65	58.11	87.34	66.29	75.64	1.18
1	54.35	36897.01	49.07	73.83	31.35	5.49	8.13	5.58	19.17	2.37
2	58.96	66197.33	48.76	505.65	284.37	49.38	68.50	49.53	72.57	2.58

3 rows × 22 columns

NumStorePurchases	NumWebVisitsMonth	AcceptedCmp1	AcceptedCmp2	AcceptedCmp3	AcceptedCmp4	AcceptedCmp5	Total Spending	Recency_Frequency	Count
8.45	3.16	0.55	0.11	0.14	0.36	0.72	1602.20	2.51	203
3.59	6.46	0.01	0.00	0.07	0.03	0.00	143.55	7.36	1102
8.41	4.21	0.02	0.00	0.07	0.07	0.00	1030.00	2.55	723

- The purpose of executing clustering techniques, specifically K-Means clustering, is to perform customer segmentation based on their behaviours (spending patterns, campaign engagement, recency of purchases, etc.). The code block involved feature engineering, feature selection (e.g., age, income, spending, purchase frequency, and campaign acceptances), and standardization. Subsequently, the elbow method determines the optimal number of clusters for K-means clustering. It computes the Within-Cluster Sum of Squares (WCSS) for 1-10 k values, and illustrates the results to check where the “elbow” starts to form. In this case, the optimal number of clusters

is 3. The K-means algorithm assigns each customer to a cluster based on their purchasing behaviour. Principal Component Analysis (PCA) is applied to reduce the dimensionality of the dataset, while preserving important relationships. The scatter plot is created using PCA components, wherein the customers are colour coded by their assigned cluster.

The clustering analysis summary reveals the following:

	Cluster 0: Premium In-store Shoppers	**Cluster 1: Budget Online Browsers**	**Cluster 2: Wine Enthusiasts**
Demographics	Middle-aged (avg. 55 years old) with high income ($80,139)	Slightly younger (54 years old) with the lowest income ($36,897)	Oldest segment (avg 59 years old) with highest income ($66,197)
Shopping Behavior	Moderate spending across categories (avg. $1,602)	Minimal spending across categories (avg. $143)	Highest overall spending ($1,030), and highly concentrated in wines ($505)
Channel Preferences	Strong preference for in-store shopping (8.45 store purchases)	Highest web visits (6.46 visits/month) but low conversion to purchases	Strong preference for in-store shopping (8.41 store purchases)
Campaign Response	Highest acceptance rates across all campaigns, especially Campaign 1 (55%) and Campaign 5 (72%).	Little to no campaign response (all under 7%)	Moderate campaign acceptance.

Classification

- The classification task in this project aims to predict campaign acceptance, wherein which customers are most likely to accept future marketing campaigns. With this prediction, it allows Nata Supermarket to allot promotional resources more efficiently, while boosting campaign success. However, it has been detected that there is an imbalance in the dataset, with fewer customer accepting campaigns than rejecting. This urged the analyst to apply the SMOTE technique. This ensures that all the machine learning models were trained on a balanced representation of 2 classes, which improves their ability to generalize.

```
Original class distribution: Target
0    1294
1     328
Name: count, dtype: int64
Balanced class distribution: Target
0    1294
1    1294
Name: count, dtype: int64
```

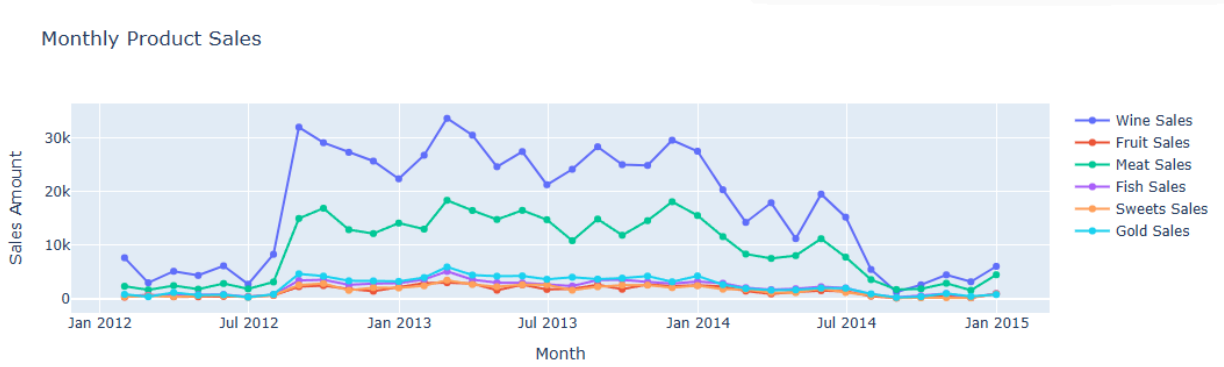
- After preprocessing the data and scaling features, four algorithms were implemented:
    1. Logistic Regression for offering insight into how features influence campaign acceptance. It is also known for its simplicity, speed, and interpretability.
    2. Random Forest enhances forecasting accuracy and stability via ensemble learning.
    3. Decision Tree is known for its rule-based logic, intuitive, and transparency on business decisions.
    4. XGBoost is a gradient boosting algorithm for its classification tasks on tabular data with high performance.
- The following models are evaluated using classification metrics. The results are summarized below:

Model	Accuracy	AUC	Precision	Recall	F1-score
Logistic Regression	0.939	0.959	0.90	0.96	0.94

Random Forest	0.958	0.969	0.93	0.97	0.96
Decision Tree	0.833	0.882	0.80	0.88	0.83
XGBoost	0.953	0.969	0.92	0.97	0.95

- Based on these results, the best performing models are Random Forest and XGBoost, wherein both achieved very high scores across all metrics. Both models are highly accurate in terms of accuracy and AUC (~96%), which means they have a strong discriminatory power between acceptors and non-acceptors With a precision that is 90% and above, and a recall of ~97%, these models are deemed to be reliable for identifying true campaign acceptors and minimizing false positives, which is critical for targeted marketing. F1-score confirms the robustness and balance of these models.
- While these outperforming models are viable, XGBoost is more preferable due to its slightly higher precision and F1-score. This balances the trade-off between seeking true acceptors and minimizing false positives. Its robustness, in terms of capability, performance, and efficiency, makes it suitable for implementation on Nata Supermarket's marketing workflow.
- Random forest is a well-performing model based on accuracy and AUC scores. However, it has slightly lower precision and recall compared to XGBoost and Random Forest. The model is strong, and it is valuable for interpretability and simplicity if needed (e.g., understanding customer behaviour through coefficients).
- Decision Tree is underperforming compared to the rest of the models, especially in accuracy and AUC scores. The recall score is moderate, yet the precision is weaker, resulting in more false positives. This could be due to overfitting, or it's less generalizable. This model is best used for quick prototypes or when transparency is valuable than performance.

Time Series Analysis



- The time series shows the monthly product sales of Nata Supermarket across the 4 years, from January 2012 to January 2015. The product categories are colour coded, including wine (dark blue), fruit (red), meat (green), fish (purple), sweets (orange), and gold (light blue). Wine and meat sales dominate the overall sales trend in terms of volume, wherein there are notable peaks between July 2012 and July 2014. Starting with the wine sales, it has the highest sales overall, peaking at around $34,000 between mid-2012 to early 2014. There is a sharp increase around the latter months of 2012, followed by fluctuating high sales. However, declining sales started in early 2014 and slightly recovered in early 2015. Meat sales, the second-highest sales category, indicate a sharp rise after mid-2012, which stabilized around $13,000-$18,000 per month. Similarly, it follows the same trend of wine sales wherein it started declining in early 2014, and afterwards, it shows a slight recovery in early 2015. On the other hand, fish, fruits, sweets, and gold sales remained consistently lower than wine and meat throughout those 4 years (~$5,000). It shows minor seasonal fluctuations, but a mostly stable trend. It peaked during late 2012 and early 2014, which aligns with high wine and meat sales. Sales gradually declined in 2014, with insignificant and less noticeable fluctuations or recovery, compared to wine and meat sales. Overall, all the product categories experience a spike in sales starting in mid-2012, followed by a decline in early 2014, and then a slight increase in early 2015.

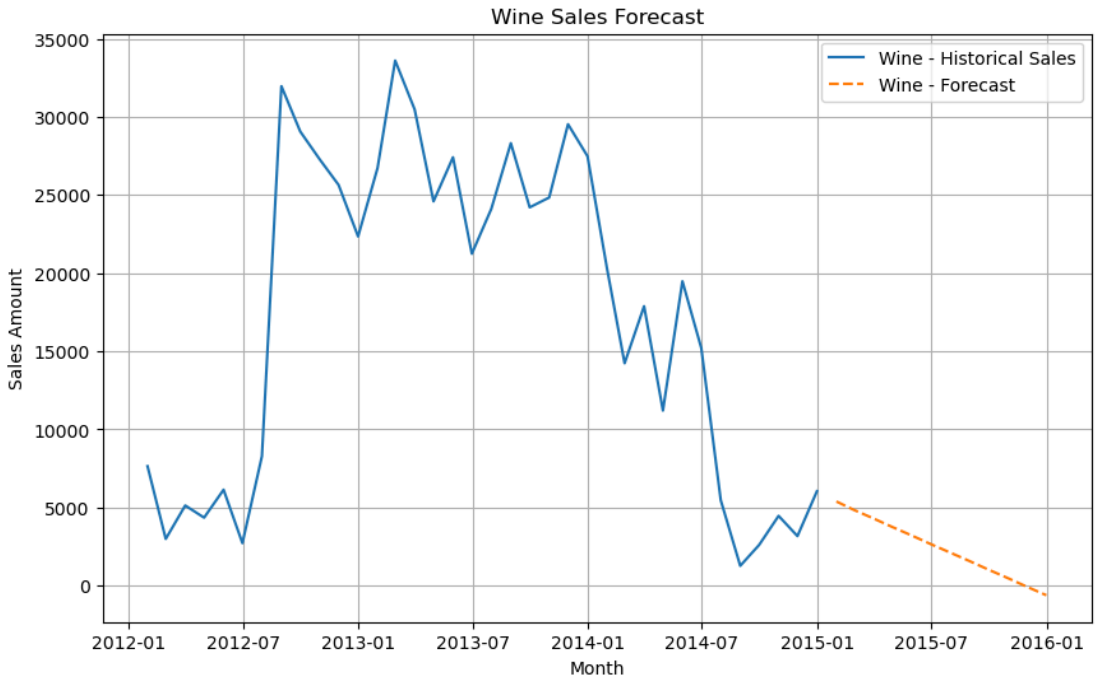
```
=== Wine Sales Stationarity Test ===
ADF Statistic: -1.654484
p-value: 0.454703
Critical Values:
        1%: -3.6327426647230316
        5%: -2.9485102040816327
        10%: -2.6130173469387756
The wine sales time series is non-stationary (p-value >= 0.05).

Second Differencing of Wine Sales:
ADF Statistic: -2.999281
p-value: 0.034962
Critical Values:
        1%: -3.7112123008648155
        5%: -2.981246804733728
        10%: -2.6300945562130176
The wine sales time series is stationary after second differencing (p-value < 0.05).

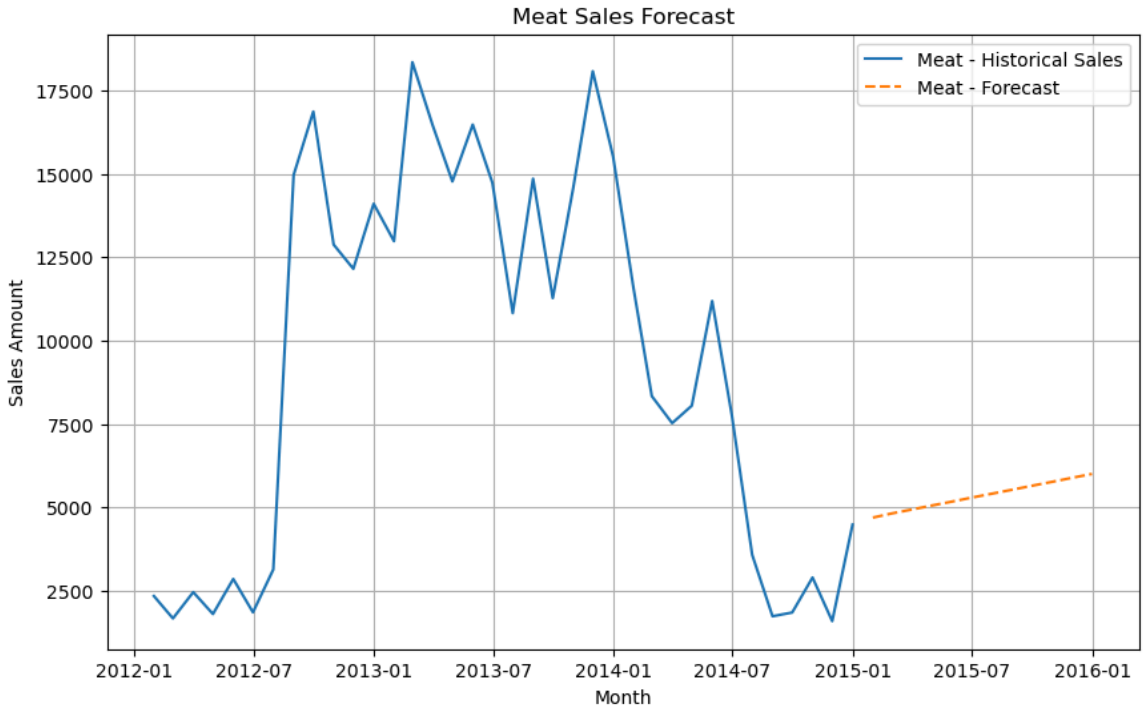

=== Meat Sales Stationarity Test ===
ADF Statistic: -1.797911
p-value: 0.381499
Critical Values:
        1%: -3.6327426647230316
        5%: -2.9485102040816327
        10%: -2.6130173469387756
The meat sales time series is non-stationary (p-value >= 0.05).

Second Differencing of Meat Sales:
ADF Statistic: -5.393218
p-value: 0.000004
Critical Values:
        1%: -3.6790595944893187
        5%: -2.9678817237279103
        10%: -2.6231583472057074
The meat sales time series is stationary after second differencing (p-value < 0.05).
```

- The purpose of this report is to test for stationarity in the wine and meat sales series using the Augmented Dickey-Fuller (ADF) test. A time series has stationarity if the shape of the distribution or statistical properties does not change when shifted in time (Alexander, 2016). Non-stationary data must be differenced before implementing time series models like ARIMA, since it assumes stationarity. When the p-value is greater than 0.05, the time series is non-stationary, which initially occurred on both wine and meat sales—the p-values are 0.455 and 0.381, respectively. Therefore, both series require second differencing to become stationary, making them ARIMA model use (d=2).



- The time series forecast for wine revealed a concerning downward trajectory and implies potential business problems ahead. The red line projects a continuing sharp decline from $5,000 to completely 0 sales by early 2016. The historical data shows that Nata Supermarket encountered major business difficulties from around 2014 onwards that severely affected sales in this category. This requires a need for immediate investigation and action.



- Unlike in the wine sales forecast, the meat forecast shows a gradual increase from early 2015 (~$5,000) to early 2016 (~$6,000). The green line represents a modest recovery, but it is not close to the previous peak levels.

Conclusion

The data science solution has revealed crucial insights that directly address the business challenges identified by Vina Verago. Through in-depth exploratory data analysis (EDA), clustering, classification, and time series analysis, the solution uncovered distinct customer groups, purchasing patterns, and alarming sales trends that require immediate attention. According to the findings, it is apparent that Nata's customer base mainly consists of middle-aged and older, married, higher-income individuals, wherein wine and meat are the widely bought product categories, driving revenue. Younger demographics, 18-24 age group, demonstrate little to no activity with the supermarket, in both sales and campaign engagement. Meanwhile, older customers (65+) have the highest spending power. The engagement on supermarket's marketing campaign portrayed significant variation wherein older and affluent customers respond better campaign promotions, particularly campaign 5.

The K-means clustering analysis has successfully grouped the customers into three distinct segments: Premium In-store Shoppers who respond actively to campaigns and have moderate spending on different categories; Budget Online Browsers are the ones who frequently visits the website, but seldomly convert to product purchases; and Wine Enthusiasts maintain the highest purchasing power with a strong preference for wine across categories. These established customer segments provide a good foundation for modifying marketing strategies that can profoundly improve campaign effectiveness and customer engagement. This helps meet the needs and preferences for each segment.

Furthermore, the classification models such as Random Forest and XGBoost (approximately 96% accuracy) are powerful solution for optimizing and strategizing future campaign targeting. Both of the models presents high recall and precision, which are capable to accurately identify customers that are more likely to accept campaigns with minimal false positives. Lastly, a serious concern is the time series analysis wherein it reveals a sharp decline in wine sales, which is also Nata's highest revenue category. This trend, along with the absence of younger age groups in the customer base, indicates potential long-term barriers for the supermarket if it is not addressed immediately. This is why the implementation of clustering and classification algorithms, along with a substantial understanding of customer behaviour and preferences, are critical to reverse these problematic trends and revitalize main product categories. It is also a valuable solution for optimizing targeted promotions, maintaining a competitive advantage, and enhancing inventory management.

Recommendation

Nata Supermarket is positioned at a crucial juncture where utilizing data science technologies has become important for maintaining a competitive edge in the supermarket industry. The company can prioritize data collection infrastructure expansion to extract more valuable insights regarding customer behaviour, inventory movement, and external factors affecting sales. This expansion of data ecosystem will allow much reliable prediction and deeper understanding of customers beyond what's currently available. To significantly enhance operational efficiency, Nata Supermarket can perhaps implement machine learning algorithms that can forecast precise inventory needs based on several factors such as demand variability, lead time, holding costs, and storage capacity (Ntsime, 2025). This will

help ensure that high-demand items like wine and meat products are adequately stocked, while reducing overstocking and waste for slow-moving goods.

Although the K-means clustering analysis provided three customer segments, further refinement and improvement can be done by integrating more behavioral data such as feedback, customer reviews, complains, and social media interactions. These data can assist in creating hyper-personalized campaigns, which ensures that customers are much engaged and receives campaigns according to their preferences. Data science techniques such as A/B testing and sentiment analysis (NLP techniques) will offer deeper analysis on which promotional strategies work the best for each customer group and adjust accordingly. Churn prediction models, loyalty program optimization, chatbots, and product recommendation engines are also some of the advanced and useful strategies that Nata Supermarket can implement to enhance customer engagement, retention, and overall business growth.

For these initiatives to succeed, Nata must invest in organisational change management, such as data literacy training and building an analytics centre of excellence. This transformation will ensure that data-driven decision-making is embedded on Nata Supermarket's culture throughout the organisation. In the company's wide expansion of data capabilities, it is notable that ethical, legal, and social considerations, in accordance with PIPEDA, must remain paramount. Transparent data policies, data anonymization or privacy, ethical AI governance framework and customer data ownership mechanisms are important to prioritize in order to build trust among customers, while safeguarding the company from regulatory challenges. Ultimately, these recommendations will position Nata Supermarket to compete effectively against Walmart and other data-driven competitors, while maintaining its pioneering commitment to customer-centred branding.

References

Kraus, S. et al. (2021) Digital Transformation: An Overview of the Current State of the Art of Research. SAGE Open, 11(3), [Online] Available from: doi.org/https://doi.org/10.1177/21582440211047576 [Accessed 23/03/2025].

Ladley, J. and Redman, T. (2020) [Online] Harvard Business Review. Available from https://hbr.org/2020/03/use-data-to-accelerate-your-business-strategy [Accessed 21/03/25].

Garg, S. (2022) Five Ways To Use Data To Elevate Customer Experience. [Online] Forbes. Available from : https://www.forbes.com/councils/forbestechcouncil/2022/03/21/five-ways-to-use-data-to-elevate-customer-experience/ [Accessed 21/03/25].

GeeksforGeeks (2024) What is Data Quality and Why is it Important? [Online] GeeksforGeeks. Available from : https://www.geeksforgeeks.org/what-is-data-quality-and-why-is-it-important/ [Accessed 23/03/25].

IBM (2024) Master data management. [Online] Ibm.com. Available from : https://www.ibm.com/think/topics/master-data-management [Accessed 23/03/25].

Kaufmann, G. (2024) The crucial role of data quality in effective data governance. [Online] Applied Data Science. Available from : https://sites.hslu.ch/applied-data-science/the-crucial-role-of-data-quality-in-effective-data-governance/ [Accessed 23/03/25].

Houghton, S. (2024) Importance of Data Security in 2025. [Online] Aztechit.co.uk. Available from : https://www.aztechit.co.uk/blog/importance-of-data-security [Accessed 23/03/25].

Dragan Sutevski, PhD (2023) The Importance of Data Security for Business: A Comprehensive Guide. [Online] Entrepreneurship in a Box. Available from : https://www.entrepreneurshipinabox.com/40240/the-importance-of-data-security-for-business-a-comprehensive-guide/ [Accessed 23/03/25].

Gonsalves, N. (2024) Ensuring Data Security Compliance: Best Practices and Strategies - AARC-360. [Online] AARC-360. Available from :

https://www.aarc-360.com/ensuring-data-security-compliance-best-practices-and-strategies/ [Accessed 23/03/25].

Avakov (2024) ResearchGate, [Online] Available from: doi.org/https://doi.org/10.13140/RG.2.2.19983.39842 [Accessed 23/03/2025].

Qualtrics (2018) What is a survey? Benefits, tips & free tool. [Online] Qualtrics. Available from : https://www.qualtrics.com/experience-management/research/survey-basics/ [Accessed 25/03/25].

Bhasin, H. (2015) What is a Customer Satisfaction Survey? Examples & Templates | Marketing91. [Online] Marketing91. Available from : https://www.marketing91.com/customer-satisfaction-questionnaire/ [Accessed 25/03/25].

Pollfish (2023) [Online] Bing. Available from : Market Research Survey: The Complete Guide - Pollfish Resources [Accessed 25/03/25].

PeopleHR (2024) [Online] Peoplehr.com. Available from : https://www.peoplehr.com/en-gb/resources/blog/employee-satisfaction-survey-questions [Accessed 25/03/25].

Ulwick (2002) [Online] Harvard Business Review. Available from : https://hbr.org/2002/01/turn-customer-input-into-innovation [Accessed 25/03/25].

Fuchs, J. (2019) What Is a Support Ticket? A Quick Definition. [Online] Hubspot.com. Available from : https://blog.hubspot.com/service/support-ticket [Accessed 25/03/25].

GeeksforGeeks (2020) What is Web Scraping and How to Use It? [Online] GeeksforGeeks. Available from : https://www.geeksforgeeks.org/what-is-web-scraping-and-how-to-use-it/ [Accessed 25/03/25].

Srivastava, S. (2023) IoT Devices and Sensors: Types, Functions, and Use Cases - E&ICT Academy, IIT Kanpur. [Online] E&ICT Academy, IIT Kanpur - E&ICT Academy, IIT Kanpur. Available from : https://eicta.iitk.ac.in/knowledge-hub/internet-of-things/iot-devices-and-sensors-types-functions-and-use-cases/ [Accessed 25/03/25].

Hosseini, M. et al. (2023) Smartwatches in healthcare medicine: assistance and monitoring; a scoping review. BMC Medical Informatics and Decision Making, 23(1), [Online] Available from: doi.org/https://doi.org/10.1186/s12911-023-02350-w [Accessed 25/03/2025].

GlobalSpec. (2015) [Online] Globalspec.com. Available from : https://insights.globalspec.com/article/1145/how-barcode-scanners-work [Accessed 25/03/25].

Amrita (2024) IoT in Retail Industry- Top 8 Benefits and Use Cases. [Online] PsiBorg Technologies Pvt. Ltd. Available from : https://psiborg.in/iot-in-retail-industry/ [Accessed 25/03/25].

Chen, J. (2024) Transaction: Definition, Accounting, and Examples. [Online] Investopedia. Available from : https://www.investopedia.com/terms/t/transaction.asp [Accessed 25/03/25].

Shopify (2022) What is a POS System? How to Choose the Right POS for 2024 - Shopify. [Online] Shopify. Available from : https://www.shopify.com/retail/point-of-sale-system [Accessed 25/03/25].

Datta, S. (2024) Financial Transaction Definition, Types & Examples. [Online] PW Skills Blog. Available from : https://pwskills.com/blog/financial-transaction [Accessed 25/03/25].

Reid, H. (2025) Understanding Inventory Movement and Its Role in Supply Chain Efficiency. [Online] DCL Logistics. Available from : https://dclcorp.com/blog/inventory/inventory-movement/ [Accessed 25/03/25].

Astera. (2020) [Online] Astera. Available from : https://www.astera.com/type/blog/data-manipulation-tools/ [Accessed 25/03/25].

Reza, M.S. (2022) Pros and Cons of Google Sheets vs Excel - Which Is Better? - ExcelDemy. [Online] ExcelDemy. Available from : https://www.exceldemy.com/pros-and-cons-of-google-sheets-vs-excel/ [Accessed 25/03/25].

Pedamkar (2018) SPSS vs SAS. [Online] EDUCBA. Available from : https://www.educba.com/spss-vs-sas/ [Accessed 25/03/25].

Mohmad (2024) R, Python, and Julia: A Comparative Study of Data Science. [Online] Analytics Insight. Available from : https://www.analyticsinsight.net/data-science/r-python-and-julia-a-comparative-study-of-data-science [Accessed 25/03/25].

IBM Cloud Education (2024) PostgreSQL vs MySQL. [Online] Ibm.com. Available from : https://www.ibm.com/think/topics/postgresql-vs-mysql [Accessed 25/03/25].

Edmond, S. and Crabtree, M. (2022) Power BI vs. Tableau: Which is The Better Business Intelligence Tool in 2025? [Online] Datacamp.com. Available from : https://www.datacamp.com/blog/power-bi-vs-tableau-which-one-should-you-choose [Accessed 25/03/25].

Venky Anant et al. (2020) The consumer-data opportunity and the privacy imperative. [Online] McKinsey & Company. Available from : https://www.mckinsey.com/capabilities/risk-and-resilience/our-insights/the-consumer-data-opportunity-and-the-privacy-imperative [Accessed 25/03/25].

Sanders, C.K. and Scanlon, E. (2021) The Digital Divide Is a Human Rights Issue: Advancing Social Inclusion Through Social Work Advocacy. Journal of Human Rights and Social Work, 6(2), pp. 130–143.

SAP (2025) [Online] Sap.com. Available from : https://www.sap.com/resources/what-is-ai-bias [Accessed 25/03/25].

Bias in AI. (2025) [Online] Chapman.edu. Available from : https://www.chapman.edu/ai/bias-in-ai.aspx[Accessed 25/03/25].

Bareebe (2022) ResearchGate, [Online] Available from: doi.org/https://doi.org/10.13140/RG.2.2.19583.69285 [Accessed 25/03/2025].

Hall, J., Kendrick, C. and Nosko, C. (2015) The Effects of Uber's Surge Pricing: A Case Study.

Sarwar, N. (2024) Overcoming Data Privacy Challenges in Digital Marketing. [Online] Webtech Solution : SEO and Digital Marketing Agency. Available from : https://www.webtechsolution.org/guide-for-overcoming-data-privacy-challenges-in-digital-marketing/ [Accessed 25/03/25].

Intersoft Consulting (2021) [Online] General Data Protection Regulation (GDPR). Available from : https://gdpr-info.eu/issues/fines-penalties [Accessed 25/03/25].

The CNIL's restricted committee imposes a financial penalty of 50 Million euros against GOOGLE LLC | European Data Protection Board. (2019) [Online] Europa.eu. Available from : https://www.edpb.europa.eu/news/national-news/2019/cnils-restricted-committee-imposes-financial-penalty-50-million-euros_en [Accessed 25/03/25].

GDPR Summary. (2020) [Online] GDPR Summary. Available from : https://www.gdprsummary.com/gdpr-summary [Accessed 25/03/25].

Masaga (2016) COMELEC data breach (2016) Case Study. ResearchGate, [Online] Available from: doi.org/https://doi.org/10.13140/RG.2.2.24128.61444 [Accessed 25/03/2025].

Office (2024) PIPEDA fair information principles - Office of the Privacy Commissioner of Canada. [Online] Priv.gc.ca. Available from : https://www.priv.gc.ca/en/privacy-topics/privacy-laws-in-canada/the-personal-information-protection-and-electronic-documents-act-pipeda/p_principle/ [Accessed 25/03/25].

Ethical Data Collection: A Guide for Business Owners. (2025) [Online] https://vivatechnology.com. Available from : https://vivatechnology.com/news/ethical-data-collection-a-guide-for-business-owners [Accessed 25/03/25].

Jimna Jayan (2024) Ethics in Data Collection: Why Responsible Web Scraping Matters. [Online] PromptCloud. Available from : https://www.promptcloud.com/blog/importance-of-ethical-data-collection/[Accessed 25/03/25].

Gargiulo, M. (2022) Data Security Threats: What You Need To Know. [Online] Forbes. Available from : https://www.forbes.com/councils/forbestechcouncil/2022/05/16/data-security-threats-what-you-need-to-know/ [Accessed 25/03/25].

Rashid, F. and Leyden, J. (2020) 9 types of phishing attacks and how to identify them. [Online] CSO Online. Available from :

https://www.csoonline.com/article/563353/8-types-of-phishing-attacks-and-how-to-identify-them.html [Accessed 25/03/25].

Danao, M. (2023) What Is A Data Breach? Definition, Examples & Prevention. [Online] Forbes Advisor. Available from : https://www.forbes.com/advisor/business/what-is-data-breach/ [Accessed 25/03/25].

NUMENTIS (2023) Numentis. [Online] NUMENTIS. Available from : https://numentis.com/password-best-practices-how-to-keep-your-digital-life-secure/ [Accessed 25/03/25].

Cisco (2018). Two-Factor Authentication (2FA) [Online] Cisco Duo. Available from : https://duo.com/product/multi-factor-authentication-mfa/two-factor-authentication-2fa [Accessed 25/03/25].

Hetler, A. (2022) 5 reasons software updates are important. [Online] WhatIs. Available from : https://www.techtarget.com/whatis/feature/5-reasons-software-updates-are-important [Accessed 27/03/25].

Luna, C. (2024) 6 Best Cybersecurity Training for Employees in 2025. [Online] eSecurity Planet. Available from : https://www.esecurityplanet.com/products/cybersecurity-training/ [Accessed 25/03/25].

paloalto (2015) IPS. vs. IDS vs. Firewall: What Are the Differences? [Online] Palo Alto Networks. Available from : https://www.paloaltonetworks.com/cyberpedia/firewall-vs-ids-vs-ips [Accessed 25/03/25].

Anh, V. (2023) The Ultimate Guide for Supermarket Management. [Online] Viindoo. Available from : https://viindoo.com/blog/business-management-3/operations-management-of-a-supermarket-1571 [Accessed 29/03/25].

Reddy, K.S. (2025) LEVERAGING BIG DATA ANALYTICS FOR SUPPLY CHAIN OPTIMIZATION IN THE FMCG SECTOR: A CASE STUDY APPROACH. International Journal of Business Intelligence Research, 3(1), pp. 1–32.

Albérico Travassos Rosário and Dias, J.C. (2023) How has data-driven marketing evolved: Challenges and opportunities with emerging technologies. International Journal of Information Management Data Insights, 3(2), pp. 100203–100203.

Basiriya and Mohammad (2019) IMPACTING CUSTOMER SATISFACTION IN SUPERMARKET. ResearchGate, [Online] Available from: doi.org/https://doi.org/10.24941/ijcr.34066.01.2019 [Accessed 29/03/2025].

Twin, A. (2024) Competitive Advantage Definition With Types and Examples. [Online] Investopedia. Available from : https://www.investopedia.com/terms/c/competitive_advantage.asp [Accessed 28/03/25].

Supermarket Strategic Plan Template | Template by ClickUpTM. (2025) [Online] Clickup.com. Available from : https://clickup.com/templates/strategic-plan/supermarket [Accessed 29/03/25].

GeeksforGeeks (2024) Data Science in Supply Chain Optimization. [Online] GeeksforGeeks. Available from : https://www.geeksforgeeks.org/data-science-in-supply-chain-optimization/ [Accessed 29/03/25].

Wingfield, T. (2023) Best Practices for Using Data to Optimize Your Supply Chain. [Online] Actian. Available from : https://www.actian.com/blog/data-analytics/best-practices-data-supply-chain/ [Accessed 29/03/25].

Thelosen, L. (2023) Drive supply chain efficiency and optimization using data and AI. [Online] Google Cloud Blog. Available from : https://cloud.google.com/blog/topics/supply-chain-logistics/drive-supply-chain-efficiency-and-optimization-using-data-and-ai [Accessed 29/03/25].

MITSDE (2024) How Data Science is Powering Personalized Marketing Strategies. [Online] MIT School of Distance Learning | Distance Learning Institute for MBA / PGDM Courses. Available from : https://blog.mitsde.com/how-data-science-is-powering-personalized-marketing-strategies/ [Accessed 29/03/25].

CONDOR (2025) What is POS (Point of Sale) | Condor PH. [Online] Condor POS Solutions. Available from : https://condorpossolutions.ph/what-is-pos/ [Accessed 02/04/25].

administrator (2025) CRM for Grocery Stores: A Complete Guide. [Online] CRM Business Tools. Available from : https://crm.teknobgt.com/crm-for-grocery-stores-a-complete-guide [Accessed 02/04/25].

Iseal (2023) Internet of Things (IoT) Applications in Grocery Retail: Smart Stores and Inventory Management. [Online] ResearchGate. Available from : https://www.researchgate.net/publication/385079814_Internet_of_Things_IoT_Applications_in_Grocery_Retail_Smart_Stores_and_Inventory_Management [Accessed 02/04/25].

Ramadan, W. (2023) Create a Database to Manage Grocery Store Data using SQL Workbench. [Online] Medium. Available from : https://medium.com/@wandiramadan.stat/create-a-database-to-manage-grocery-store-data-using-sql-workbench-ec351cf2e6d7 [Accessed 02/04/25].

Langdoc (2023) How the grocery industry can innovate and transform with AWS | Amazon Web Services. [Online] Amazon Web Services. Available from : https://aws.amazon.com/blogs/industries/how-the-grocery-industry-can-innovate-and-transform-with-aws/ [Accessed 02/04/25].

Mounika, D. et al. (2022) Predictive Analysis of Supermarket Sales Using Machine Learning. International Journal of Creative Research Thoughts, 10(6), pp. 2320–2882.

Vikramaditya Naruka (2024) Mastering Time Series Forecasting: Predicting Grocery Store Sales with Machine Learning. [Online] Medium. Available from : https://medium.com/@vikramadityanaruka/mastering-time-series-forecasting-predicting-grocery-store-sales-with-machine-learning-067b3dd9aa00 [Accessed 02/04/25].

John, J.M., Shobayo, O. and Ogunleye, B. (2023) An Exploration of Clustering Algorithms for Customer Segmentation in the UK Retail Market. Analytics, 2(4), pp. 809–823.

Retail Analytics | Tableau. (2025) [Online] Tableau. Available from : https://www.tableau.com/solutions/industries/retail-consumer-goods/retail[Accessed 02/04/25].

Team, C. (2025) Inventory Turnover. [Online] Corporate Finance Institute. Available from : https://corporatefinanceinstitute.com/resources/accounting/inventory-turnover/ [Accessed 02/04/25].

Vipond, T. (2024) Price Elasticity. [Online] Corporate Finance Institute. Available from : https://corporatefinanceinstitute.com/resources/economics/price-elasticity/[Accessed 02/04/25].

GeeksforGeeks (2024) 7 Common Data Science Challenges and Effective Solutions. [Online] GeeksforGeeks. Available from : https://www.geeksforgeeks.org/7-common-data-science-challenges-and-effective-solutions/ [Accessed 02/04/25].

Shalomova, A. (2023) What Are Resource Constraints And How To Manage Them? [Online] PPM Express. Available from : https://ppm.express/blog/resource-constraints/ [Accessed 02/04/25].

Lesego Ntsime (2025) Inventory Levels: Types, Strategies, and How to Calculate Them. [Online] Intuendi - Orchestration and automation for your entire supply chain. Available from : https://intuendi.com/resource-center/inventory-levels/ [Accessed 06/04/25].